

Feldman–Hájek, but Non-Gaussian: Geometry of the Weighted Fourier Jet

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Abstract

Kakutani’s theorem tensorizes equivalence of product measures, but a bounded invertible linear mixing operator T destroys the independence on which that test rests. The resulting difficulty is genuinely infinite-dimensional: every fixed finite output marginal may look regular while the full image law is singular. For a non-Gaussian i.i.d. product whose coordinate law has an a.e.-positive density, finite second moment, and finite location and scale Fisher information, we prove that equivalence, Hilbert–Schmidt proximity to the exact linear stabilizer, and bounded projected Hellinger energy are equivalent. The weighted Fourier jet supplies finite-frequency, dimension-free Hellinger witnesses that turn local mixing into Frobenius cost. We also classify standard symmetric α -stable products for $0 < \alpha < 2$. In this infinite-variance branch, the exact cost is anisotropic: diagonal deviations are square summable, while the ℓ^2 -norms of the off-diagonal input columns are α -summable. Thus even an everywhere-positive coordinate density does not make Hilbert–Schmidt proximity sufficient. When a finite-variance density has zeros, the Hilbert–Schmidt criterion has an exact extension by the zero-set row condition: the rows of both T and T^{-1} must avoid the zero set. Half-line and bounded supports then reduce the equivalence group to explicit permutation classes.

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1 Introduction and Main Results

1.1 The question and answer

Fix one non-Gaussian coordinate law ρ and the i.i.d. product $\mu = \rho^{\otimes \mathbb{N}}$. For which bounded invertible linear maps T on ℓ^2 is the induced image law equivalent to μ ? Kakutani’s product criterion answers coordinatewise perturbations, but it does not survive bounded invertible linear mixing: the transformed coordinates cease to be independent and the Hellinger affinity no longer tensorizes. The obstruction is not visible from any fixed finite window, since every fixed finite output marginal can have an a.e.-positive density even when the full image law is singular. Under the finite-variance and Fisher hypotheses of the main theorem, the continuous Gaussian symmetry collapses to the allowed signed coordinate permutations, and the surviving continuous thickness is exactly Hilbert–Schmidt. Equivalently, the projected Hellinger energies of the finite output marginals are bounded exactly on that equivalence branch. This conclusion is not a consequence of the a.e.-positive-density assumption alone. For standard symmetric α -stable coordinates with $0 < \alpha < 2$, the density is everywhere positive but the exact criterion is governed by an anisotropic mixing cost: diagonal errors are square summable and the off-diagonal input-column norms are α -summable. When a finite-variance density satisfies the Sobolev assumptions but has zeros, the Hilbert–Schmidt condition must be supplemented by the zero-set row condition: the rows of both T and T^{-1} must avoid the zero set of the one-dimensional density.

1.2 Historical context: from Kakutani to bounded invertible linear mixing

Kakutani’s theorem turns equivalence of infinite products into a scalar summability problem. The test is the product of one-coordinate Hellinger affinities, positive exactly on the equivalence branch and zero exactly on the singular branch [10]. Sharp criteria for sequences of Euclidean and, more generally, Lie-group perturbations retain this product structure and identify Fisher information as the local Hellinger metric [4, 15].

The Gaussian law conceals the obstruction created by bounded invertible linear mixing. For the standard Gaussian product, every orthogonal operator on ℓ^2 has a measurable row-series realization and preserves the product law. Translation and linear measure-class changes are governed by the Cameron–Martin and Feldman–Hájek theories, with explicit Gaussian density criteria developed by Shepp and nonlinear extensions by Ramer [2, 3, 5, 7, 12, 14]. A non-Gaussian product behaves differently: exact independence rigidifies the orthogonal group to coordinate permutations and to those coordinate signs already preserving the one-dimensional law. Non-Gaussian quasi-invariant measures on Hilbert and linear topological spaces have long been known [6, 16]; the problem here is the complete measure-class classification for bounded invertible linear mixing of one fixed i.i.d. product law.

1.3 The phase transition: continuous Gaussian symmetry and discrete rigidity

Let X have a centered probability law ρ with finite first moment and characteristic function $\phi(u) = \mathbb{E}e^{iuX}$. The rotational Fourier defect used below has a specific job: it turns the dependence created by a small non-Gaussian rotation into a bounded observable whose strength does not deteriorate with the ambient dimension. Define the rotational Fourier defect by

$$R(\xi, \eta) = \eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta). \quad (1)$$

It is the first variation of the expectation of a product of centered complex exponentials under a planar rotation. Its determinant representation uses the weighted Fourier jet, the explicit curve

$$u \longmapsto (u\phi(u), \phi'(u)) \in \mathbb{C}^2. \quad (2)$$

Up to sign, $R(\xi, \eta)$ is the determinant of the two vectors on this curve at ξ and η . Hence zero rotational Fourier defect means that the curve lies in one fixed complex line through the origin. The resulting global linear ODE has only Gaussian characteristic functions as solutions.

Under the finite-second-moment and Sobolev assumptions below, a finite set of frequencies detects every infinitesimal matrix direction for a non-Gaussian coordinate law. The corresponding Hellinger orbit is uniformly coercive in Frobenius norm near the exact linear stabilizer, independently of dimension. In that finite-variance regime, the surviving continuous thickness is exactly Hilbert–Schmidt: equivalence holds precisely for Hilbert–Schmidt perturbations of coordinate permutations whose signs preserve ρ . The symmetric α -stable regime has the same exact linear stabilizer but a different, anisotropic accumulation law, as shown in [Section 4](#).

1.4 Main theorem: equivalence, projected Hellinger energy, and the exact linear stabilizer

The main finite-variance theorem has a simple nontechnical content. For coordinate laws satisfying (3), bounded invertible linear mixing preserves the product measure class exactly when it differs by Hilbert–Schmidt size from the exact linear stabilizer. The projected Hellinger

energies defined below are not heuristic proxies for this condition; their boundedness is an equivalent diagnostic. An everywhere-positive density alone does not imply this Hilbert–Schmidt criterion: the standard symmetric α -stable laws have everywhere positive densities, but their infinite-variance classification requires the stronger anisotropic mixing cost in (74).

We now state the assumptions used for the theorem on bounded invertible linear mixing. Suppose $\rho(dx) = f(x) dx$, put $g = \sqrt{f}$, and assume

$$\begin{aligned} f > 0 \quad \text{Lebesgue-almost everywhere,} \quad g &\in W^{1,2}(\mathbb{R}), \quad xg' \in L^2(\mathbb{R}), \\ \mathbb{E}X &= 0, \quad \mathbb{E}X^2 = 1, \quad f \neq (2\pi)^{-1/2} e^{-x^2/2}. \end{aligned} \quad (3)$$

All derivatives are weak derivatives. Cusps are allowed. No moment of order greater than two is assumed. The two Sobolev conditions are finite location and scale Fisher information.

Put $H = \ell^2(\mathbb{N}; \mathbb{R})$ and $\mu = \rho^{\otimes \mathbb{N}}$ on the product Borel space $\mathbb{R}^{\mathbb{N}}$, and $\mu_n = \rho^{\otimes n}$. The space H indexes coefficients. Write $\mathcal{B}(H)$ for its bounded linear operators, $\text{GL}(H)$ for the invertible members, and $(e_i)_{i \geq 1}$ for the standard coordinate basis. A sample from μ is not asserted to belong to H . Define

$$\begin{aligned} \mathcal{P} = \{Q : (Qx)_i &= \varepsilon_i x_{\pi(i)}, \quad \pi \text{ a permutation of } \mathbb{N}, \\ \varepsilon_i &\in \{-1, 1\}, \quad \text{Law}(\varepsilon_i X) = \rho \text{ for every } i\}. \end{aligned} \quad (4)$$

Write $\mathcal{P}^\pm$ for the full signed-permutation group. Thus $\mathcal{P} = \mathcal{P}^\pm$ when ρ is symmetric, whereas $\mathcal{P}$ is the ordinary permutation group when ρ is not symmetric. This definition also applies when ρ has no density. For a bounded linear operator T on H , let Φ_T be the measurable row-series realization constructed in lemma 3.1. Write $\nu_T = (\Phi_T)_\# \mu$.

For probability measures λ, ν , set

$$A(\lambda, \nu) = \int \sqrt{\frac{d\lambda}{dm} \frac{d\nu}{dm}} dm, \quad h(\lambda, \nu)^2 = 2 - 2A(\lambda, \nu), \quad (5)$$

where m dominates both measures. We use the same notation for densities and their probability measures. Let $P_n : H \rightarrow H$ be the orthogonal projection onto the first n coordinate vectors, and let $\pi_n : \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}^n$ extract the first n coordinates. Write $\mathcal{S}_2(H)$ for the Hilbert–Schmidt ideal. Define the projected Hellinger energy by

$$E_n(T) = -2 \log A(\mu_n, (\pi_n)_\# \nu_T). \quad (6)$$

These are actual output marginals, not laws obtained from principal matrix corners. Thus E_n keeps the finite evidence produced by the actual output rows and asks whether that evidence remains uniformly bounded as the window grows.

Theorem 1.1 (Complete criterion for bounded invertible linear mixing). *Under (3), for every $T \in \text{GL}(H)$,*

$$\boxed{\begin{aligned} \nu_T \sim \mu &\iff \exists Q \in \mathcal{P} : TQ^{-1} - I \in \mathcal{S}_2(H) \\ &\iff \sup_n E_n(T) < \infty. \end{aligned}} \quad (7)$$

If these conditions fail, then $\nu_T \perp \mu$ and $E_n(T) \uparrow \infty$. The exact linear stabilizer in $\text{GL}(H)$ is $\mathcal{P}$: $\nu_T = \mu$ if and only if $T \in \mathcal{P}$.

Proposition 1.2 (Finite-coordinate approximants recover the likelihood ratio). *Under the assumptions of [theorem 1.1](#), suppose $\nu_T \sim \mu$. For any admissible Q , put $K = TQ^{-1} - I$ and $T_n = I + P_n K P_n$. Then T_n is eventually invertible and*

$$\frac{d\nu_{T_n}}{d\mu} \longrightarrow \frac{d\nu_{TQ^{-1}}}{d\mu} \quad \text{in } L^1(\mu), \quad \sqrt{\frac{d\nu_{T_n}}{d\mu}} \longrightarrow \sqrt{\frac{d\nu_{TQ^{-1}}}{d\mu}} \quad \text{in } L^2(\mu). \quad (8)$$

Since $\nu_Q = \mu$, the limiting law is also ν_T . Moreover, the finite-marginal likelihood ratios and their reciprocals are uniformly integrable in their respective reference measures.

[Proposition 1.2](#) is the analytic bookkeeping behind the criterion: finite-dimensional likelihood computations do not merely approximate cylinder probabilities, but converge to the actual infinite-dimensional likelihood on the equivalence branch.

The theorem contains the global energy formulation through $\sup_n E_n(T)$. The rotational Fourier defect is not an additional scalar condition on T . It is the calibration mechanism. It turns small Hellinger displacement into squared Frobenius displacement. This distinction keeps the one-dimensional Gaussian characterization separate from the operator-valued equivalence criterion.

Two refinements identify the assumptions needed for the stabilizer and for orthogonal transformations.

Corollary 1.3 (Moment-only stabilizer). *If ρ is non-Gaussian and centered with variance one, then the exact linear stabilizer of $\mu = \rho^{\otimes \mathbb{N}}$ is $\mathcal{P}$. No density or Fisher information assumption is needed.*

Corollary 1.4 (Equivalence under orthogonal transformations). *If $f > 0$ almost everywhere, $\sqrt{f} \in W^{1,2}(\mathbb{R})$, and f is non-Gaussian and centered with variance one, then every orthogonal U obeys*

$$\nu_U \sim \mu \iff U - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}.$$

Thus scale Fisher information is needed for bounded invertible linear mixing but not for the orthogonal subproblem.

1.5 Result map

The classification separates metric accumulation from the zero-set row condition. The following map records the strongest conclusion in each regime and the reason that conclusion matters.

Condition	Strongest conclusion	Value
Finite-variance assumptions (3); bounded invertible linear mixing	$\nu_T \sim \mu \iff \exists Q \in \mathcal{P} : TQ^{-1} - I \in \mathcal{S}_2 \iff \sup_n E_n(T) < \infty$	A complete and checkable measure-class boundary for bounded invertible linear mixing.
Non-Gaussian a.e.-positive density centered with variance one and with finite location Fisher information; orthogonal mixing	The same Hilbert–Schmidt criterion, assuming only location Fisher information.	The important rotation regime needs weaker regularity.
Standard symmetric α -stable coordinates, $0 < \alpha < 2$; compatible stable row-series realization	Equivalence holds exactly when, after an allowed signed permutation, the diagonal deviations are square summable and the off-diagonal input-column norms are α -summable; equivalently, the stable projected Hellinger energies stay bounded.	Diagonal errors have quadratic cost, while off-diagonal input columns have $\ell^\alpha(\ell^2)$ cost; Hilbert–Schmidt proximity alone is not sufficient.
Finite-variance Sobolev assumptions (109); density with zeros	Hilbert–Schmidt proximity plus the forward and inverse zero-set row conditions.	Positive affinity is separated from true two-sided absolute continuity.
Half-line or bounded support	The equivalence group collapses to explicit permutation or signed-permutation classes.	Support boundaries can eliminate every bounded invertible linear mixing beyond signed coordinate permutations.

1.6 Organization and notation

Section 2 develops the Fourier detector, the matrix Fisher geometry, and the dimension-free Hellinger bounds. Section 3 constructs the measurable row action and proves the finite-variance classification, including the exact linear stabilizer and global coercivity. Section 4 classifies standard symmetric α -stable products and identifies their anisotropic mixing cost, with diagonal-square and off-diagonal-column terms. Section 5 gives the exact extension for densities with zeros, proves the bounded-support and half-line classifications, and separates the roles of the finite-variance hypotheses.

We write A^* for the adjoint, including the transpose of a real matrix, and $\|\cdot\|_{\text{op}}$ for the operator norm. The Frobenius inner product is $\langle A, B \rangle_F = \text{tr}(A^*B)$, with norm $\|A\|_F$. The law ρ remains fixed unless a different law is explicitly specified.

Symbol	Meaning
ρ, μ, μ_n	coordinate law and its infinite and n -coordinate products
ϕ	characteristic function $u \mapsto \mathbb{E}e^{iuX}$
R	rotational Fourier defect in (1)
A, h	Hellinger affinity and Hellinger distance
P_n, π_n	orthogonal projection on H and coordinate extraction on $\mathbb{R}^{\mathbb{N}}$
$E_n(T)$	projected Hellinger energy in (6)
$\mathcal{S}_2$	Hilbert–Schmidt ideal; $\ \cdot\ _{\mathcal{S}_2} = \ \cdot\ _F$ in finite dimensions
$\mathcal{P}, \mathcal{P}^{\pm}$	coordinate permutations with signs preserving ρ , and the full signed-permutation group
Φ_T, ν_T	measurable row-series realization and its pushforward of μ

The notation for the symmetric stable branch is introduced at its first use in [Section 4](#).

2 Rotational Fourier Defect and Hellinger Coercivity

Non-Gaussianity is first detected by a bounded, logarithm-free rotational Fourier defect: if it vanishes at every frequency pair, a global linear ODE forces the coordinate law to be Gaussian [\[8, 9\]](#). The matrix Fisher operator then identifies the infinitesimal geometry, and finite real Fourier sums turn a nonzero defect into dimension-free dual tests. This yields local Frobenius coercivity. We also state the global form modulo the exact linear stabilizer; its proof follows the infinite-dimensional classification.

2.1 A product of centered complex exponentials

This subsection requires only $\mathbb{E}X = 0$ and $\mathbb{E}|X| < \infty$. Neither a density nor a second moment is needed. Write $\phi(u) = \mathbb{E}e^{iuX}$, and for independent copies X_1, X_2 put

$$O_{\theta} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad W_{\xi, \eta}(x, y) = (e^{i\xi x} - \phi(\xi))(e^{i\eta y} - \phi(\eta)).$$

This fixes the sign convention for rotation. Centering removes the two marginal responses. It leaves only the dependence created by rotation. In particular,

$$\mathbb{E}W_{\xi, \eta}(X_1, X_2) = 0, \quad \mathbb{E}|W_{\xi, \eta}(X_1, X_2)|^2 = (1 - |\phi(\xi)|^2)(1 - |\phi(\eta)|^2) \leq 1. \quad (9)$$

Pointwise, $|W_{\xi, \eta}| \leq 4$, so its variance under an arbitrary comparison law is at most 16. Both bounds are dimension-free and require no tail assumption.

2.2 Gaussian characterization by a global linear ODE

The next lemma makes the rotational Fourier defect a logarithm-free rigidity tool, valid even when ϕ has zeros. Contrapositively, every non-Gaussian law has a nonzero rotational response.

Lemma 2.1 (Gaussian characterization by the rotational Fourier defect). *For the product of centered complex exponentials defined above,*

$$R(\xi, \eta) = \frac{d}{d\theta} \mathbb{E} W_{\xi, \eta}(O_\theta(X_1, X_2)) \Big|_{\theta=0} = \eta \phi'(\xi) \phi(\eta) - \xi \phi(\xi) \phi'(\eta). \quad (10)$$

Moreover,

$$R \equiv 0 \iff \text{Law}(X) \text{ is a centered Gaussian, possibly degenerate.} \quad (11)$$

Equivalently, all pairs of values of the weighted Fourier jet

$$u \mapsto (u\phi(u), \phi'(u))$$

are linearly dependent if and only if $\text{Law}(X)$ is a centered Gaussian, possibly degenerate.

Proof. The first moment implies $\phi \in C^1(\mathbb{R})$. The joint characteristic function after rotation is

$$\phi(\xi \cos \theta + \eta \sin \theta) \phi(-\xi \sin \theta + \eta \cos \theta).$$

The marginal characteristic functions are $\phi(\xi \cos \theta) \phi(-\xi \sin \theta)$ and $\phi(\eta \sin \theta) \phi(\eta \cos \theta)$. Their derivatives at zero vanish because $\phi'(0) = i\mathbb{E}X = 0$. Differentiating the joint characteristic function therefore gives (10).

If R vanishes everywhere, continuity and $\phi(0) = 1$ allow us to choose $\eta_0 \neq 0$ with $\phi(\eta_0) \neq 0$. Fix this single frequency. The defect equation then becomes the global ODE

$$\phi'(\xi) = c\xi\phi(\xi), \quad c = \frac{\phi'(\eta_0)}{\eta_0\phi(\eta_0)}, \quad \phi(0) = 1.$$

No division by $\phi(\xi)$ has occurred. Uniqueness for this linear ODE gives $\phi(\xi) = \exp(c\xi^2/2)$ on all of $\mathbb{R}$, including any putative zero of ϕ . The identity $\phi(-\xi) = \overline{\phi(\xi)}$ forces $c \in \mathbb{R}$. Also $|\phi(\xi)| \leq 1$ forces $c \leq 0$. Write $c = -\sigma^2$. The Gaussian conclusion follows. If $\sigma = 0$, we get the point mass at zero. Substitution proves the converse. $\square$

2.3 The determinant representation and finite detection

The determinant form supplies the finite frequency pair used to construct the dual moments in [Section 2.7](#).

Consider the curve

$$u \mapsto (u\phi(u), \phi'(u)) \in \mathbb{C}^2,$$

described as the weighted Fourier jet. Its values satisfy

$$\det \begin{pmatrix} \xi\phi(\xi) & \phi'(\xi) \\ \eta\phi(\eta) & \phi'(\eta) \end{pmatrix} = -R(\xi, \eta). \quad (12)$$

The weighting by u is essential: for a nondegenerate Gaussian law, the unweighted curve $u \mapsto (\phi(u), \phi'(u))$ is not contained in a fixed line.

For a non-Gaussian law, continuity supplies distinct frequencies and a compact detecting neighborhood, independent of the ambient dimension; the diagonal $\xi = \eta$ never detects the law.

The same product of centered complex exponentials gives a local Hellinger lower bound for one rotation. Choose a phase α with $\operatorname{Re}(e^{-i\alpha}R(\xi, \eta)) > 0$ and apply

$$|\mathbb{E}_\lambda F - \mathbb{E}_\nu F| \leq h(\lambda, \nu) \{2\mathbb{E}_\lambda F^2 + 2\mathbb{E}_\nu F^2\}^{1/2}$$

to $F = \operatorname{Re}(e^{-i\alpha}W_{\xi, \eta})$. It follows that $h((O_\theta)_\# \operatorname{Law}(X)^{\otimes 2}, \operatorname{Law}(X)^{\otimes 2}) \geq c|\theta|$ for small $|\theta|$. A first moment suffices for this lower bound. The dimension-free matrix estimates below require the additional second-moment and Sobolev hypotheses.

2.4 Dimension-free variance and affine normalization

The dimension-free variance bounds needed below are already contained in (9); we record only how R changes under affine normalization.

Centering, scaling, and phase. Let Y have mean m and characteristic function ϕ_Y , and put $X = (Y - m)/\sigma$, where $\sigma > 0$. For $a = \xi/\sigma$ and $b = \eta/\sigma$, the rotational Fourier defect of X satisfies

$$R(\xi, \eta) = e^{-i(m/\sigma)(\xi+\eta)} [b\phi'_Y(a)\phi_Y(b) - a\phi_Y(a)\phi'_Y(b) + im(a-b)\phi_Y(a)\phi_Y(b)]. \quad (13)$$

The bracketed term is the mean-corrected rotational response for two copies of Y . Centering contributes the displayed phase, scaling changes the frequencies, and the modulus at corresponding frequencies is unchanged. For a centered symmetric law the defect is real, but its phase at a single pair is not a complete measure of skewness.

For the concrete comparison $Y \sim \text{Gamma}(2, 1)$, $m = 2$, $\sigma = \sqrt{2}$, and $(\xi, \eta) = (1, 2)$, the expression in brackets in (13) equals $4/27$, so the law centered with variance one has

$$R(1, 2) = \frac{4}{27} e^{-3i\sqrt{2}}.$$

Thus the defect can remain explicit even for an asymmetric law with a support boundary.

2.5 The matrix Fisher operator

The matrix Fisher operator separates diagonal scale, symmetric off-diagonal, and skew-symmetric rotational directions. Its rotational eigenvalue is $J - 1$, which vanishes exactly in the Gaussian case.

Let f be a probability density and write $g = \sqrt{f}$. In this subsection we assume

$$\begin{aligned} f > 0 \quad & \text{Lebesgue-almost everywhere,} \quad g \in W^{1,2}(\mathbb{R}), \quad xg'(x) \in L^2(\mathbb{R}), \\ \mathbb{E}X &= 0, \quad \mathbb{E}X^2 = 1, \quad X \sim f(x) dx. \end{aligned} \quad (14)$$

The derivatives in these assumptions are weak derivatives. Define the location score $s = -f'/f = -2g'/g$ almost everywhere. Then

$$\mathbb{E}s(X) = 0, \quad \mathbb{E}[Xs(X)] = 1, \quad J := \mathbb{E}s(X)^2 = 4\|g'\|_2^2. \quad (15)$$

Define the scale score and its second moment by

$$d(x) := xs(x) - 1, \quad \kappa := \mathbb{E}d(X)^2 = 4\|xg'\|_2^2 - 1. \quad (16)$$

The two integration-by-parts identities in (15) follow from these assumptions. Indeed, $f' = 2gg'$ belongs to L^1 , and xf' belongs to L^1 by Cauchy–Schwarz and the finite second moment. Choose a smooth compactly supported cutoff χ equal to one near zero. Applying integration by parts first to $\chi(x/L)$ and then to $x\chi(x/L)$ gives the identities as $L \rightarrow \infty$.

For independent copies $X_1, \dots, X_n$ of X , write $\mathbf{X}^{(n)} = (X_1, \dots, X_n) \sim \mu_n$. For $A \in M_n(\mathbb{R})$, define the score of the pushforward by $I + tA$ at $t = 0$ by

$$S_A(\mathbf{X}^{(n)}) = \sum_{i,j=1}^n A_{ij}(s(X_i)X_j - \delta_{ij}) = \sum_i A_{ii}d(X_i) + \sum_{i \neq j} A_{ij}s(X_i)X_j. \quad (17)$$

Theorem 2.2 (The matrix Fisher operator). *Under (14), let s and d be the location and scale scores defined above, and define the matrix Fisher operator $\mathcal{I}$ by $\langle A, \mathcal{I}B \rangle_F = \mathbb{E}[S_A S_B]$. Then*

$$\boxed{\mathcal{I}A = JA + A^* + (\kappa - J - 1) \text{diag } A.} \quad (18)$$

Here $\text{diag } A$ is the matrix with diagonal entries A_{ii} and zero off-diagonal entries. The diagonal, symmetric off-diagonal, and skew-symmetric matrix subspaces are orthogonal for the Frobenius inner product and have eigenvalues κ , $J + 1$, and $J - 1$, respectively. Equivalently,

$$\begin{aligned} \langle A, \mathcal{I}A \rangle_F &= \kappa \|\text{diag } A\|_F^2 + (J + 1) \left\| \frac{A + A^*}{2} - \text{diag } A \right\|_F^2 \\ &\quad + (J - 1) \left\| \frac{A - A^*}{2} \right\|_F^2. \end{aligned} \quad (19)$$

The eigenvalue multiplicities are, respectively,

$$n, \quad n(n - 1)/2, \quad n(n - 1)/2. \quad (20)$$

Moreover, $\kappa > 0$ and $J \geq 1$. Here $J = 1$ holds precisely for the standard Gaussian density. Thus, for a non-Gaussian f , $\min\{\kappa, J - 1\} > 0$ and

$$\begin{aligned} \min\{\kappa, J - 1\} \|A\|_F^2 &\leq \langle A, \mathcal{I}A \rangle_F \leq C_f \|A\|_F^2, \\ C_f &:= \max\{\kappa, J + 1\}. \end{aligned} \quad (21)$$

Proof. The diagonal variables $d(X_i)$ are independent and centered. Their covariances with every off-diagonal term $s(X_j)X_k$, $j \neq k$, vanish. After any shared coordinate is integrated out, the remaining factor has expectation $\mathbb{E}X$ or $\mathbb{E}s(X)$. Two off-diagonal terms can have a nonzero covariance only when their ordered index pairs coincide or are reversed. Those covariances are, respectively, J and $(\mathbb{E}Xs(X))^2 = 1$. Consequently

$$\mathbb{E}S_A^2 = \kappa \sum_i A_{ii}^2 + \sum_{i < j} \{J(A_{ij}^2 + A_{ji}^2) + 2A_{ij}A_{ji}\}. \quad (22)$$

Resolving each pair into its symmetric and skew-symmetric parts proves the decomposition. In particular, the calculation uses centering and independence and imposes no parity condition on f .

Cauchy–Schwarz applied to $\mathbb{E}Xs(X) = 1$ gives $J \geq 1$. Equality implies $s(X) = X$ almost surely, hence $g' = -xg/2$ almost everywhere and f is the standard Gaussian. If $\kappa = 0$, then $2xg' + g = 0$ almost everywhere on each of the two half-lines. The absolutely continuous solutions there are constant multiples of $|x|^{-1/2}$, which cannot give a positive probability density. This proves $\kappa > 0$ and the asserted bounds. $\square$

2.6 Hellinger paths, polar decomposition, and symmetric perturbations

For orientation, write the left polar decomposition of an invertible matrix as $T = BO$, where $B = (TT^*)^{1/2}$ is positive and O is orthogonal. The covariance defect is $TT^* - I = B^2 - I$, so a finite second moment detects the symmetric factor. The skew tangent of the orthogonal factor has Fisher eigenvalue $J - 1$, which is positive precisely off the Gaussian law. The full matrix Fisher operator decomposition above packages these two mechanisms without requiring a nonlinear separation of B and O at every step.

For $A \in \text{GL}(n, \mathbb{R})$ define a unitary operator on $L^2(\mathbb{R}^n, dx)$ by

$$(U_A v)(y) = |\det A|^{-1/2} v(A^{-1}y), \quad \Omega_n(x) = \prod_{i=1}^n g(x_i).$$

Thus $U_A \Omega_n$ is the square root of the Lebesgue density of $A_{\#} \mu_n$. For invertible A, B ,

$$h(A_{\#} \mu_n, B_{\#} \mu_n) = \|U_A \Omega_n - U_B \Omega_n\|_2.$$

For a fixed real matrix A , the generator of $U_{\exp(tA)}$ is

$$G_A v = -\frac{1}{2} \text{tr}(A)v - (Ax) \cdot \nabla v, \quad G_A \Omega_n = \frac{1}{2} S_A \Omega_n. \quad (23)$$

These identities hold in L^2 . To justify them under (14), observe that $x_j \partial_i \Omega_n \in L^2$ for all i, j : the diagonal case uses $xg' \in L^2$, and the off-diagonal case uses $g' \in L^2$ and $\mathbb{E}X^2 = 1$. Cutoff and mollification approximate Ω_n in the corresponding weighted Sobolev graph norm, where the usual change-of-variables differentiation is valid.

Lemma 2.3 (A dimension-free Hellinger path bound). *If A_t , $0 \leq t \leq 1$, is a continuously differentiable path in $\text{GL}(n, \mathbb{R})$, then $t \mapsto U_{A_t} \Omega_n$ is continuously differentiable in L^2 and*

$$\begin{aligned} \frac{d}{dt} U_{A_t} \Omega_n &= U_{A_t} G_{A_t^{-1} \dot{A}_t} \Omega_n, \\ \left\| \frac{d}{dt} U_{A_t} \Omega_n \right\|_2 &= \frac{1}{2} \sqrt{\left\langle A_t^{-1} \dot{A}_t, \mathcal{I}(A_t^{-1} \dot{A}_t) \right\rangle_F}. \end{aligned} \quad (24)$$

Consequently,

$$h((A_1)_{\#} \mu_n, (A_0)_{\#} \mu_n) \leq \frac{\sqrt{C_f}}{2} \int_0^1 \|A_t^{-1} \dot{A}_t\|_F dt. \quad (25)$$

In particular, if $\|B\|_{\text{op}} \leq r < 1$, then

$$h((I + B)_{\#} \mu_n, \mu_n) \leq \frac{\sqrt{C_f}}{2(1-r)} \|B\|_F. \quad (26)$$

All constants are independent of n .

Proof. The representation law $U_A U_B = U_{AB}$ and (23) give (24). Integrate the derivative and use the Fisher upper bound. For the last assertion take $A_t = I + tB$ and use $\|A_t^{-1}\|_{\text{op}} \leq (1-r)^{-1}$. $\square$

2.7 Finite real Fourier sums with dual moments

Assume in addition that f is non-Gaussian. The next lemma turns the rank-two obstruction from [lemma 2.1](#) into bounded dual witnesses: finite real combinations of $1, \cos(\xi x), \sin(\xi x)$, at frequencies depending on f but not on the matrix dimension.

Lemma 2.4 (Finite real Fourier sums with dual moments). *There exist centered finite real Fourier sums p, q, r such that*

$$\begin{array}{c|cc} & s & x \\ \hline p & 1 & 0 \\ q & 0 & 1 \end{array} \quad \text{and} \quad \mathbb{E}[r(X)d(X)] = 1. \quad (27)$$

The entries in the table denote expectations of the products. For

$$H_A(\mathbf{X}^{(n)}) := \sum_i A_{ii}r(X_i) + \sum_{i \neq j} A_{ij}p(X_i)q(X_j), \quad (28)$$

one has, for all real matrices A, B of the same size,

$$\mathbb{E}[H_A S_B] = \langle A, B \rangle_F, \quad \|H_A\|_{L^2(\mu_n)} \leq C_{\text{dual}} \|A\|_F, \quad (29)$$

where C_{dual} is independent of n .

Proof. On the real vector space of centered finite real Fourier sums consider the two functionals

$$L_1(u) = \mathbb{E}u'(X) = \mathbb{E}[u(X)s(X)], \quad L_2(u) = \mathbb{E}[Xu(X)].$$

The integration-by-parts identity holds since u, u' are bounded. Both functionals are unchanged by subtracting a constant. If this moment map had rank at most one, all its two-by-two determinants would vanish, including after complexification. Writing $\phi(\xi) = \mathbb{E}e^{i\xi X}$ and testing with the centered exponentials at frequencies ξ, η would then give

$$\eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta) = 0 \quad (\xi, \eta \in \mathbb{R}).$$

Here $L_1(e^{i\xi x}) = i\xi\phi(\xi)$ and $L_2(e^{i\xi x}) = -i\phi'(\xi)$. The global linear ODE argument in [lemma 2.1](#) would force a centered Gaussian law, a contradiction. The map therefore has rank two. Inverting the moment matrix of two finite real Fourier sums supplies p, q with the displayed dual moments. This construction fixes finitely many frequencies depending only on f .

The diagonal function can be chosen using only the second moment. For $\xi \neq 0$ let $r_0(x) = \cos(\xi x) - \mathbb{E}\cos(\xi X)$. Dominated convergence gives

$$\frac{\mathbb{E}[Xr'_0(X)]}{\xi^2} = -\mathbb{E}\left[X\frac{\sin(\xi X)}{\xi}\right] \rightarrow -\mathbb{E}X^2 = -1 \quad (\xi \rightarrow 0).$$

Choose a sufficiently small nonzero ξ and set $r = r_0/\mathbb{E}[Xr'_0(X)]$. Cutoff integration by parts gives $\mathbb{E}[r(X)d(X)] = \mathbb{E}[Xr'(X)] = 1$.

All diagonal/off-diagonal cross terms in $\mathbb{E}H_A S_B$ vanish by centering. Coincident ordered off-diagonal pairs contribute $\mathbb{E}[ps]\mathbb{E}[qX] = 1$. Reversed pairs contribute $\mathbb{E}[pX]\mathbb{E}[qs] = 0$. This proves the pairing. For any centered $a, b \in L^2(f)$, independence gives

$$\left\| \sum_{i \neq j} A_{ij}a(X_i)b(X_j) \right\|_2 \leq \sqrt{2} \|a\|_2 \|b\|_2 \|A\|_F. \quad (30)$$

Together with the independent diagonal terms, this proves the last assertion. $\square$

The functions H_A will be paired with the Hellinger tangent along a matrix path. To keep that pairing uniformly controlled after moving away from the identity, we need one uniform derivative bound for $H_A\Omega_n$. The next lemma is the technical price of keeping the test functions finite-frequency and dimension-free. Its proof gives the index contractions explicitly and uses no fourth moment of X or higher moment of the score.

Lemma 2.5 (Generator bound for the functions H_A). *Let p, q, r be any fixed centered bounded continuously differentiable functions with bounded derivatives. Define H_A by (28). There is a constant C_{gen} , depending on these functions and on J, κ , such that*

$$\|G_B(H_A\Omega_n)\|_{L^2(dx)} \leq C_{\text{gen}}\|A\|_F\|B\|_F \quad (A, B \in M_n(\mathbb{R}), n \geq 1). \quad (31)$$

Proof. For fixed n , both H_A and its first derivatives are bounded. Consequently

$$x_j\partial_i(H_A\Omega_n) = H_Ax_j\partial_i\Omega_n + x_j(\partial_iH_A)\Omega_n \in L^2(dx)$$

by the weighted Sobolev regularity of Ω_n and $\mathbb{E}X^2 = 1$. The cutoff and mollification argument used for (23) therefore puts $H_A\Omega_n$ in the domain of G_B . The product rule gives

$$G_B(H_A\Omega_n) = \left\{ \frac{1}{2}H_AS_B - (Bx) \cdot \nabla H_A \right\} \Omega_n. \quad (32)$$

Both terms belong to L^2 in each finite dimension. We prove a dimension-free bound for each term.

We will repeatedly use the following elementary consequence of independence. If $u_1, \dots, u_m$ are centered $L^2(f)$ functions, and a coefficient tensor C is indexed by ordered tuples of *distinct* coordinates, then

$$\left\| \sum_{i_1, \dots, i_m}^{\neq} C_{i_1 \dots i_m} \prod_{a=1}^m u_a(X_{i_a}) \right\|_2 \leq \sqrt{m!} \|C\|_{\ell^2} \prod_{a=1}^m \|u_a\|_2. \quad (33)$$

In the square of this norm, a mixed term vanishes unless the two tuples have the same index set. There are at most $m!$ matching permutations. Cauchy–Schwarz for the local moments gives a bound on each index pattern. A second Cauchy–Schwarz step for the coefficient arrays proves the full bound. We use it only for $m \leq 4$.

First consider H_AS_B . For a vector a and a zero-diagonal matrix M , abbreviate

$$D_a(u) = \sum_i a_i u(X_i), \quad Q_M(u, v) = \sum_{i \neq j} M_{ij} u(X_i) v(X_j).$$

The product H_AS_B is the sum of the four products obtained by multiplying $D_{\mathbf{a}}(r) + Q_{A_0}(p, q)$ and $D_{\mathbf{b}}(d) + Q_{B_0}(s, x)$, where $\mathbf{a} = (A_{ii})_i$, $\mathbf{b} = (B_{ii})_i$, $A_0 = A - \text{diag } A$, and $B_0 = B - \text{diag } B$. In every common coordinate, a local product is a bounded function times one of d, s, x and hence is in $L^2(f)$. Write each such local product as its mean plus its centered part.

For a product $D_a(u)D_b(v)$, distinct indices give a coefficient tensor bounded by $\|a\|_2\|b\|_2$; equal indices give the Hadamard vector $(a_i b_i)_i$ and the scalar $\sum_i a_i b_i$. Their norms have the same bound. For $D_a(u)Q_M(v, w)$, there are three index patterns: three distinct indices, equality with the first index of Q_M , and equality with its second index. The first gives the tensor $a \otimes M$. In either equality case, centering the local product gives a matrix obtained by row or column

multiplication of M by a , and its mean gives M^*a or Ma . Each norm is at most $\|a\|_2\|M\|_F$. The product $Q_M(u, v)D_a(w)$ has exactly the same three patterns and bounds.

For clarity, all patterns in a quadratic–quadratic product can also be enumerated. Consider

$$Q_M(u, v)Q_N(w, z) = \sum_{i \neq j, k \neq l} M_{ij}N_{kl}u_i v_j w_k z_l,$$

where u, v are bounded and w, z are in L^2 . Four distinct indices give the restricted tensor $M \otimes N$, of norm at most $\|M\|_F\|N\|_F$. There are four patterns with exactly one shared index: $i = k$, $i = l$, $j = k$, and $j = l$. For example, the pattern $i = k$ is

$$\sum_{i, j, l}^{\neq} M_{ij}N_{il}(uw)_i v_j z_l.$$

Its centered local product has coefficient norm bounded by

$$\left\{ \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_l N_{il}^2 \right) \right\}^{1/2} \leq \|M\|_F \|N\|_F.$$

Its local mean has coefficient matrix $(M^*N)_{jl}$ off the diagonal; excluding $i = j, l$ does not change the sum because M, N have zero diagonal. The other three shared-index patterns give the same estimate with a transpose or with the two coordinates exchanged, and their mean contractions are matrix products of M, N or their transposes.

Finally, there are two patterns with both indices shared: $(k, l) = (i, j)$ and $(k, l) = (j, i)$. Their coefficients are $M_{ij}N_{ij}$ and $M_{ij}N_{ji}$. After centering the two local products, the resulting coefficients are a Hadamard matrix, its row sums, its column sums, and its total sum. For instance

$$\sum_i \left| \sum_j M_{ij}N_{ij} \right|^2 \leq \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_j N_{ij}^2 \right) \leq \|M\|_F^2 \|N\|_F^2,$$

and the other three estimates follow from the same Cauchy–Schwarz inequality. These cases exhaust the possible index coincidences. Applying (33) to the centered pieces therefore proves

$$\|H_A S_B\|_2 \leq C \|A\|_F \|B\|_F. \quad (34)$$

It remains to bound $(Bx) \cdot \nabla H_A$. We give the contractions separately, since derivatives of a centered function need not be centered. For the diagonal part the sum is

$$\sum_{i, k} A_{ii} B_{ik} r'(X_i) X_k.$$

When $i \neq k$, split r' into its mean and centered part. The latter has coefficient matrix $(\text{diag } A)B$, with its diagonal deleted. The mean has coefficient vector $B^* \mathbf{a} - \mathbf{a} \odot \mathbf{b}$. Their norms are bounded by, respectively, one and two times $\|A\|_F \|B\|_F$. When $i = k$, center the local function $xr'(x)$, which is in L^2 since r' is bounded. The coefficient vector and the mean are bounded by the same product of matrix norms.

Let $M = A_0$. The derivative of the off-diagonal part is the sum of

$$F(M, B; p', q) := \sum_{i \neq j, k} M_{ij} B_{ik} p'(X_i) q(X_j) X_k$$

and $F(M^*, B; q', p)$. In F , first take k distinct from i, j and split p' into its mean and centered part. The centered part has three-index coefficient norm bounded by

$$\left\{ \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_k B_{ik}^2 \right) \right\}^{1/2} \leq \|M\|_F \|B\|_F.$$

The mean part has, off the diagonal, the matrix $M^*B - M^* \text{diag } B$; the second term removes $i = k$, and $i = j$ already contributes zero. Its Frobenius norm is at most $2\|M\|_F \|B\|_F$. When $k = i$, center $xp'(x)$; the resulting matrix has entries $B_{ii}M_{ij}$ and its mean has vector $M^*\mathbf{b}$. When $k = j$, split both p' and $xq(x)$ into means and centered parts; the coefficients are the Hadamard matrix $M \odot B$, its row and column sums, and its total sum. The preceding Hadamard estimates bound all of them. All the local functions used here belong to L^2 , since p', q are bounded and $\mathbb{E}X^2 = 1$. The same proof applies to $F(M^*, B; q', p)$. Using (33) again proves

$$\|(Bx) \cdot \nabla H_A\|_2 \leq C\|A\|_F \|B\|_F.$$

Together with (34) and (32), this proves the lemma. $\square$

We can now close the local argument. The dual Fourier functions give a linear functional that sees the first-order displacement, while the generator bound controls the error made along the exponential path. This turns the nonzero rotational Fourier defect into a uniform Hellinger lower bound for every small matrix perturbation.

Proposition 2.6 (Uniform local Hellinger coercivity). *For every non-Gaussian density satisfying (14), there are constants $c, C, \delta > 0$, independent of n , such that*

$$c\|B\|_F \leq h((I + B)_{\#}\mu_n, \mu_n) \leq C\|B\|_F \quad \text{whenever } \|B\|_F \leq \delta. \quad (35)$$

The lower bound can be obtained using the fixed finite set of coordinate frequencies from lemma 2.4.

Proof. Fix the dual functions from lemma 2.4. First use exponential coordinates. Unitarity and the one-parameter group derivative give

$$\langle H_A \Omega_n, (U_{\exp A} - I) \Omega_n \rangle = \int_0^1 \langle U_{\exp(-tA)}(H_A \Omega_n), G_A \Omega_n \rangle dt.$$

Replacing $U_{\exp(-tA)}(H_A \Omega_n)$ by $H_A \Omega_n$ in the integral gives the leading term $\frac{1}{2} \mathbb{E} H_A S_A = \frac{1}{2} \|A\|_F^2$. For a vector in the domain of G_A , the fundamental theorem of calculus gives $\|(U_{\exp(-tA)} - I)v\|_2 \leq t\|G_A v\|_2$. Thus lemma 2.5 and the Fisher upper bound imply

$$\left| \langle H_A \Omega_n, (U_{\exp A} - I) \Omega_n \rangle - \frac{1}{2} \|A\|_F^2 \right| \leq \frac{C_{\text{gen}} \sqrt{C_f}}{4} \|A\|_F^3.$$

Cauchy–Schwarz and (29) now give, for all sufficiently small $\|A\|_F$,

$$h((\exp A)_{\#}\mu_n, \mu_n) \geq \frac{1}{4C_{\text{dual}}} \|A\|_F.$$

For $\|B\|_{\text{op}} < 1/2$, the real power-series logarithm $A = \log(I + B)$ exists and satisfies

$$\|A - B\|_F \leq \sum_{k \geq 2} \frac{\|B\|_{\text{op}}^{k-1}}{k} \|B\|_F \leq C\|B\|_{\text{op}} \|B\|_F.$$

Its norm is therefore uniformly comparable to $\|B\|_F$ in a fixed small chart. This proves the lower bound for $I + B$; the upper bound follows from lemma 2.3. $\square$

The half-density pairing supplies the uniform local certificate; tail symmetry in [Section 3.5](#) applies it to arbitrary output marginals without a separate mixed-law variance bound for H_A .

2.8 Uniform coercivity modulo the exact linear stabilizer

The following global form is proved in [Section 3.8](#), after the infinite-dimensional classification supplies the compactness input; it is not used in that classification.

Theorem 2.7 (Dimension-free global Hellinger coercivity). *Assume (3). Let $\mathcal{P}_n$ denote the allowed signed-permutation matrices in dimension n , and set*

$$d_n(T) = \min_{Q \in \mathcal{P}_n} \|T - Q\|_F. \quad (36)$$

For every $0 < a \leq b < \infty$ there are constants $c = c(f, a, b) > 0$ and $C = C(f, a, b) < \infty$ such that, for every n and every $T \in \text{GL}(n, \mathbb{R})$ whose singular values lie in $[a, b]$,

$$c \min\{d_n(T)^2, 1\} \leq h(\mu_n, T_{\#}\mu_n)^2 \leq C \min\{d_n(T)^2, 1\}. \quad (37)$$

3 Finite-Variance Equivalence and Global Rigidity

Because the product sample space is $\mathbb{R}^{\mathbb{N}}$, not ℓ^2 , we first realize bounded operators by compatible measurable row series, establish the orbit dichotomy, and recover the full affinity from finite marginals. We then prove the finite-variance classification: Hilbert–Schmidt perturbations yield convergent likelihood ratios, whereas nonsummable tail deviations produce a Hellinger obstruction. The classification finally supplies the compactness input for global finite-dimensional coercivity.

3.1 Measurable row-series realizations on countable operator orbits

Only $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$ are needed here. Since a sample from $\mu = \rho^{\otimes \mathbb{N}}$ need not lie in ℓ^2 , the row action and its inverse, tail exchanges, and likelihood cocycles require compatible measurable versions.

Lemma 3.1 (Measurable row-series realization on each countable operator orbit). *For every $T \in \mathcal{B}(H)$, there is a measurable row-series realization, namely a Borel map $\Phi_T : \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}^{\mathbb{N}}$ whose coordinates are the $L^2(\mu)$ row sums*

$$(\Phi_T x)_i = \sum_{j \geq 1} T_{ij} x_j.$$

These maps can be chosen so that the following statements hold. If a probability measure λ satisfies, for some finite $C_\lambda \geq 0$,

$$\int \left| \sum_j a_j x_j \right|^2 d\lambda(x) \leq C_\lambda \|a\|_2^2 \quad \text{for every finitely supported } a, \quad (38)$$

then $(\Phi_T)_{\#}\lambda$ also satisfies this condition. Its constant is at most $C_\lambda \|T\|_{\text{op}}^2$. Also

$$\Phi_S \circ \Phi_T = \Phi_{ST} \quad \lambda\text{-almost surely} \quad (S, T \in \mathcal{B}(H)). \quad (39)$$

For each fixed countable subgroup $\Gamma \leq \text{GL}(H)$, there is a Γ -invariant Borel set $D \subset \mathbb{R}^{\mathbb{N}}$ of full measure for every λ satisfying (38). On this set, these restrictions form an action by bimeasurable automorphisms. In particular, this applies to μ and all its Γ -orbit laws, including singular orbit laws.

Proof. For $a \in \ell^2$, the finite sums $\sum_{j \leq n} a_j x_j$ have a unique $L^2(\lambda)$ limit L_a , by (38). Choose

$$n_k(a) = \min \left\{ n \geq k : \sum_{j > n} |a_j|^2 \leq 2^{-4k} \right\}.$$

Define $\widehat{L}_a(x)$ to be the limit of $\sum_{j \leq n_k(a)} a_j x_j$ when it exists, and zero otherwise. The estimate

$$\left\| \sum_{j \leq n_k(a)} a_j x_j - L_a \right\|_{L^2(\lambda)}^2 \leq C_\lambda 2^{-4k}$$

and Borel–Cantelli show that $\widehat{L}_a = L_a$, λ -almost surely. This one definition therefore works for every law satisfying (38); it does not depend on knowing whether that law is equivalent to μ . Set $\Phi_T(x)_i = \widehat{L}_{T^*e_i}(x)$. The construction is Borel, and the same is true jointly in the matrix coefficients of T and in x , since each tail norm and the integer choice n_k is Borel.

For finitely supported b , the finite sum of output coordinates equals L_{T^*b} almost surely. This proves the asserted second-moment bound for the pushforward. For the i -th row of S , finite output sums satisfy

$$\sum_{j \leq n} S_{ij} \widehat{L}_{T^*e_j} = L_{T^*P_n S^*e_i} \longrightarrow L_{T^*S^*e_i} \quad \text{in } L^2(\lambda).$$

The selected subsequence on the left also converges almost surely to $\Phi_S(\Phi_T(x))_i$, because the output law has the required second-moment bound. Uniqueness of the limit proves (39).

For a countable Γ , let B be the Borel set on which all its row limits, identity relations, and pairwise composition relations hold. The preceding argument gives $\lambda(B) = 1$ for every law under consideration. Put

$$D = \bigcap_{T \in \Gamma} \Phi_T^{-1}(B).$$

Every orbit law satisfies the same second-moment condition, so $\lambda(D) = 1$. The composition identities imply that D is invariant and that $\Phi_{T^{-1}}$ is the inverse of Φ_T there. This proves the final assertion. $\square$

Write $\nu_T = (\Phi_T)_\# \mu$. Every countable operator family used below therefore acts on one common carrier. No universal pointwise action of all $\text{GL}(H)$ is needed; on the original product law these versions agree with the usual almost-sure row sums.

3.2 Rational translation invariance and the zero–one dichotomy

Quasi-invariance under ℓ^2 -translations eliminates the intermediate case: every bounded invertible orbit is either equivalent to μ or singular to it.

Lemma 3.2 (Dichotomy for every bounded invertible orbit). *For every $T \in \text{GL}(H)$,*

$$\nu_T \sim \mu \quad \text{or} \quad \nu_T \perp \mu. \quad (40)$$

This conclusion uses only the a.e. positivity of f , its centered finite second moment, and $\sqrt{f} \in W^{1,2}(\mathbb{R})$. In particular, it holds under (3).

Proof. Put $g = \sqrt{f}$. The translation estimate

$$1 - A(f, f(\cdot - a)) = \frac{1}{2} \|g(\cdot - a) - g\|_2^2 \leq \frac{1}{2} \|g'\|_2^2 a^2$$

and Kakutani's theorem imply $(x \mapsto x + a)_{\#} \mu \sim \mu$ for every $a \in \ell^2$. The row realizations satisfy

$$\Phi_T(x + T^{-1}a) = \Phi_T(x) + a \quad \mu\text{-almost surely};$$

this follows directly by adding the absolutely convergent scalar row pairing with the ℓ^2 vector $T^{-1}a$. Consequently, ν_T is also quasi-invariant under every ℓ^2 translation.

Let $\mathcal{A}$ be the countable additive group generated by $\{qe_j, qTe_j : q \in \mathbb{Q}, j \geq 1\}$. Both μ and ν_T are quasi-invariant and ergodic under translation by $\mathcal{A}$. Indeed, an event invariant modulo μ under all rational finite-coordinate translations has, by a.e. positivity of the finite-dimensional densities, almost-everywhere constant sections in each finite coordinate block. It is therefore a tail event modulo μ , and has probability zero or one. Here one uses the elementary fact that a measurable function on $\mathbb{R}^n$ invariant almost everywhere under rational translations is constant almost everywhere, obtained, for example, by convolution with smooth approximate identities. For ν_T , pull an invariant event back by Φ_T ; invariance under qTe_j becomes invariance under qe_j , giving the same zero-one conclusion.

Write $\nu_T = \nu_a + \nu_s$ for its Lebesgue decomposition relative to μ . Choose a Borel μ -null set N_0 supporting ν_s , and let $N = \bigcup_{a \in \mathcal{A}} (N_0 + a)$. Quasi-invariance of μ makes N a μ -null set; it is exactly $\mathcal{A}$ -invariant and still supports ν_s . Thus $\nu_T(N) = \nu_s(\mathbb{R}^{\mathbb{N}})$, and ergodicity of ν_T makes this number zero or one. The latter case gives singularity. In the former case, write $\nu_T = \ell_T \mu$. Quasi-invariance of both measures implies that $\{\ell_T > 0\}$ is invariant modulo μ under $\mathcal{A}$. Its μ -measure is positive because $\int \ell_T d\mu = 1$, and ergodicity of μ makes that measure one. This gives equivalence. $\square$

3.3 Finite-marginal likelihood ratios and projected Hellinger energy

Finite-marginal likelihood ratios form martingales and their affinities decrease to the full affinity. Consequently bounded projected Hellinger energy is an exact equivalence diagnostic, not merely a necessary condition.

Lemma 3.3 (Projected Hellinger energy and finite-marginal likelihood ratios). *Assume $f > 0$ almost everywhere, $\sqrt{f} \in W^{1,2}(\mathbb{R})$, and $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$. For $T \in \text{GL}(H)$, every finite output law $(\pi_n)_{\#} \nu_T$ is equivalent to μ_n . Writing $\mathcal{F}_n = \sigma(x_1, \dots, x_n)$, define*

$$\begin{aligned} \ell_{T,n}(x) &= \frac{d[(\pi_n)_{\#} \nu_T]}{d\mu_n}(\pi_n x), & E_n(T) &= -2 \log \mathbb{E}_{\mu} \sqrt{\ell_{T,n}}, \\ A(\mu_n, (\pi_n)_{\#} \nu_T) &= \mathbb{E}_{\mu} \sqrt{\ell_{T,n}}. \end{aligned} \quad (41)$$

Then $(\ell_{T,n})_{n \geq 1}$ is a nonnegative mean-one μ -martingale. We have

$$A(\mu_n, (\pi_n)_{\#} \nu_T) \downarrow A(\mu, \nu_T), \quad \nu_T \sim \mu \iff \sup_n E_n(T) < \infty. \quad (42)$$

Otherwise $E_n(T) \uparrow \infty$. Then $\nu_T \perp \mu$. If $\nu_T \sim \mu$ with density $\ell_T = d\nu_T/d\mu$, then

$$\ell_{T,n} = \mathbb{E}_\mu[\ell_T \mid \mathcal{F}_n] \longrightarrow \ell_T \quad \text{in } L^1(\mu), \quad (43)$$

$$\ell_{T,n}^{-1} = \mathbb{E}_{\nu_T}[\ell_T^{-1} \mid \mathcal{F}_n] \longrightarrow \ell_T^{-1} \quad \text{in } L^1(\nu_T). \quad (44)$$

In particular, both likelihood-ratio martingales are uniformly integrable in their respective reference measures.

Proof. The first n rows of T are linearly independent, since T^* is injective. Some n input columns therefore form an invertible $n \times n$ minor A . Split the input into these coordinates and their complement. The first n outputs have the form $AU + W$, where $U \sim \mu_n$ and W is independent of U . Conditioning on W writes their density as a mixture of translates of the a.e.-positive density of AU . This mixture is positive and finite almost everywhere, proving the finite-dimensional equivalence.

Consistency of projected laws proves the martingale assertion. The martingale limit is $\ell_{T,\infty} = d(\nu_T)_a/d\mu$, the density of the absolutely continuous part of ν_T . To see the identification explicitly, Fatou's lemma on cylinder sets gives $\ell_{T,\infty}\mu \leq \nu_T$, first on the cylinder algebra and then on the generated sigma-field. Conversely, if $(\nu_T)_a = \ell_{ac}\mu$, then $\mathbb{E}_\mu[\ell_{ac} \mid \mathcal{F}_n] \leq \ell_{T,n}$, so martingale convergence gives $\ell_{ac} \leq \ell_{T,\infty}$. The two inequalities identify the limit. Since $\mathbb{E}_\mu(\sqrt{\ell_{T,n}})^2 = 1$, the square roots are uniformly integrable in $L^1(\mu)$. They thus converge in L^1 to $\sqrt{\ell_{T,\infty}}$. Conditional Jensen's inequality gives the monotonicity of their expectations, and consequently

$$\lim_n \mathbf{A}(\mu_n, (\pi_n)_\# \nu_T) = \int \sqrt{\ell_{T,\infty}} d\mu = \mathbf{A}(\mu, \nu_T).$$

The dichotomy in [lemma 3.2](#) now proves (42) and the singular alternative.

Under equivalence, the forward conditional-expectation identity is the definition of a finite-marginal density. The reverse identity follows because for $B \in \mathcal{F}_n$,

$$\int_B \ell_{T,n}^{-1} d\nu_T = \mu(B) = \int_B \ell_T^{-1} d\nu_T.$$

Conditional-expectation convergence gives both L^1 limits and hence both uniform-integrability assertions. $\square$

Because initial segments are cofinal among finite coordinate sets, data processing also shows that the criterion is independent of the coordinate enumeration.

3.4 Sufficiency of Hilbert–Schmidt perturbations

Finite-coordinate Hilbert–Schmidt approximants are Cauchy in Hellinger distance, and the proposition identifies their limit as an infinite-dimensional likelihood ratio.

Proposition 3.4 (Hilbert–Schmidt sufficiency and likelihood-ratio convergence). *Assume (14), let $\mu = \rho^{\otimes \mathbb{N}}$, where $\rho(dx) = f(x) dx$, and let $T = I + K \in \text{GL}(H)$ with $K \in \mathcal{S}_2$. Then $\nu_T \sim \mu$. For the eventually invertible operators $T_n = I + P_n K P_n$, their likelihood ratios $\ell_{T_n} = d\nu_{T_n}/d\mu$ converge in $L^1(\mu)$ to the strictly positive likelihood ratio $\ell_T = d\nu_T/d\mu$, and $\sqrt{\ell_{T_n}} \rightarrow \sqrt{\ell_T}$ in $L^2(\mu)$. The inverse approximants satisfy the same conclusion:*

$$\ell_{T_n^{-1}} \longrightarrow \ell_{T^{-1}} \quad \text{in } L^1(\mu).$$

For $S, T \in \text{GL}(H)$ with $S - I, T - I \in \mathcal{S}_2$, the likelihood ratios obey, on the common measurable carrier of their countable operator group,

$$\ell_{ST}(y) = \ell_S(y) \ell_T(\Phi_{S^{-1}} y), \quad (45)$$

$$\ell_T(y) \ell_{T^{-1}}(\Phi_{T^{-1}} y) = 1. \quad (46)$$

If Q is an exact linear stabilizer of μ , the same equivalence holds whenever $TQ^{-1} - I \in \mathcal{S}_2$ and T is invertible.

Proof. Since $P_n K P_n \rightarrow K$ in Hilbert–Schmidt norm, $T_n \rightarrow T$ in operator norm. The approximants are therefore eventually invertible. Their inverse norms are uniformly bounded. If A_n is the first n -coordinate block of T_n , then

$$\ell_{T_n}(x) = |\det A_n|^{-1} \prod_{i=1}^n \frac{f((A_n^{-1} \pi_n x)_i)}{f(x_i)}. \quad (47)$$

These densities are positive almost everywhere. For all sufficiently large m, n , the operator $T_n^{-1} T_m - I$ acts in a finite coordinate block and has operator norm at most $1/2$. Hellinger invariance under the finite invertible map T_n^{-1} and (26) give

$$\|\sqrt{\ell_{T_m}} - \sqrt{\ell_{T_n}}\|_2 \leq \sqrt{C_f} \|T_n^{-1}\|_{\text{op}} \|T_m - T_n\|_{\mathcal{S}_2}. \quad (48)$$

Thus the square roots are Cauchy in $L^2(\mu)$. They have a nonnegative limit of norm one; denote its square by $\tilde{\ell}_T$. Cauchy–Schwarz also gives

$$\|\ell_{T_m} - \ell_{T_n}\|_1 \leq 2 \|\sqrt{\ell_{T_m}} - \sqrt{\ell_{T_n}}\|_2, \quad \ell_{T_n} \longrightarrow \tilde{\ell}_T \quad \text{in } L^1(\mu). \quad (49)$$

Let $\mathbf{X} = (X_j)_{j \geq 1}$ have law μ . For each fixed finite set of output coordinates, the row-series realizations $\Phi_{T_n} \mathbf{X}$ converge to $\Phi_T \mathbf{X}$ in L^2 . Bounded continuous cylinder tests identify $\tilde{\ell}_T \mu$ with ν_T .

The inverse approximants converge in Hilbert–Schmidt norm because

$$\|T_n^{-1} - T^{-1}\|_{\mathcal{S}_2} \leq \|T_n^{-1}\|_{\text{op}} \|T_n - T\|_{\mathcal{S}_2} \|T^{-1}\|_{\text{op}}.$$

Apply the same argument to these finite-coordinate approximants. This gives $\nu_{T^{-1}} \ll \mu$. The measurable row-series construction in lemma 3.1 requires only the centered finite second moment. On its countable-group carrier it gives

$$\mu = (\Phi_T)_{\#} \nu_{T^{-1}} \ll (\Phi_T)_{\#} \mu = \nu_T.$$

This proves equivalence and identifies $\tilde{\ell}_T = \ell_T > 0$ almost everywhere. Change of variables on the same carrier proves the cocycle and inverse identities. Finally, $\nu_Q = \mu$ and $T = (TQ^{-1})Q$ give the assertion about the exact linear stabilizer. $\square$

3.4.1 The orthogonal case requires only location Fisher information

For orthogonal transformations the assumptions can be reduced to

$$f > 0 \text{ almost everywhere,} \quad g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \quad \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1. \quad (50)$$

In particular, $xg' \in L^2$ is not required. The cutoff identities $\mathbb{E}s(X) = 0$ and $\mathbb{E}[Xs(X)] = 1$ still follow from $J < \infty$ and the second moment. For a skew-symmetric matrix A , the score has only off-diagonal terms and

$$\mathbb{E}S_A^2 = (J - 1)\|A\|_F^2. \quad (51)$$

The generator formula remains valid. Only $x_j \partial_i \Omega_n$ with $i \neq j$ occur. One can justify it by radial cutoff and radial mollification. Those operators commute with the orthogonal one-parameter group.

Proposition 3.5 (Orthogonal local bounds and Hilbert–Schmidt sufficiency). *Under (50), every orthogonal U on ℓ^2 with $U - I \in \mathcal{S}_2$ satisfies $\nu_U \sim \mu$. Its likelihood ratios have the same L^1 and square-root L^2 convergence along finite-coordinate orthogonal approximants, as do the inverse likelihood ratios. If f is non-Gaussian, there also exist dimension-free $c, C, \delta > 0$ such that*

$$c\|V - I\|_F \leq h(V_{\#}\mu_n, \mu_n) \leq C\|V - I\|_F \quad (V \in O(n), \|V - I\|_F \leq \delta). \quad (52)$$

Proof. For a skew-symmetric A , integration along $\exp(tA)$ gives

$$h((\exp A)_{\#}\mu_n, \mu_n) \leq \frac{1}{2}\sqrt{J - 1}\|A\|_F. \quad (53)$$

If f is non-Gaussian, the construction of the two dual functions p, q in lemma 2.4 uses only $s(X), X \in L^2$ and $J - 1 > 0$. On skew-symmetric matrices use $H_A = \sum_{i \neq j} A_{ij} p(X_i) q(X_j)$, with no diagonal function. The score S_B likewise has no diagonal term. Consequently the proof of lemma 2.5 contains only bounded functions times $s(X)$ or X , and proves the same estimate under (50). The half-density pairing argument gives the local lower bound in exponential coordinates. For an orthogonal V sufficiently close to I , its power-series logarithm is real and skew-symmetric, with $\|\log V\|_F$ comparable to $\|V - I\|_F$. This proves (52).

For completeness, finite-coordinate orthogonal approximants can be constructed without applying a nonorthogonal path estimate. Set $T_n = I + P_n(U - I)P_n$ and, for large n , take its polar factor

$$V_n = T_n(T_n^* T_n)^{-1/2}.$$

It is orthogonal and acts identically outside the first n coordinates. Since $T_n \rightarrow U$ in Hilbert–Schmidt norm,

$$\|T_n^* T_n - I\|_{\mathcal{S}_2} \leq (\|T_n\|_{\text{op}} + 1)\|T_n - U\|_{\mathcal{S}_2} \rightarrow 0.$$

The smallest singular values of T_n are bounded below by some $a > 0$. Functional calculus for $|T_n|$ then gives

$$\|V_n - T_n\|_{\mathcal{S}_2} = \|I - |T_n|\|_{\mathcal{S}_2} \leq \frac{\|T_n^* T_n - I\|_{\mathcal{S}_2}}{1 + a}.$$

Thus $V_n \rightarrow U$ in Hilbert–Schmidt norm. For sufficiently large n, m , $V_n^* V_m$ is close to the identity and has a small skew-symmetric logarithm. Applying (53) to this logarithm proves that the square roots of the likelihood ratios are Cauchy. The L^1 limits of the likelihood ratios are identified by the same cylinder tests as in proposition 3.4. Apply the argument also to $V_n^{-1} \rightarrow U^{-1}$; the measurable inverse action then proves equivalence and the asserted convergence. $\square$

3.5 Necessity of Hilbert–Schmidt perturbations

Approximation and likelihood-ratio convergence transfer local coercivity from coordinate blocks to Hilbert–Schmidt operators, allowing many small tail mismatches to be combined into a contradiction.

Lemma 3.6 (The local bound for Hilbert–Schmidt operators). *Let f be non-Gaussian and satisfy (14). There are constants $c, C, \delta > 0$ such that every $K \in \mathcal{S}_2(\ell^2)$ with $\|K\|_{\mathcal{S}_2} \leq \delta$ satisfies*

$$c\|K\|_{\mathcal{S}_2} \leq h(\nu_{I+K}, \mu) \leq C\|K\|_{\mathcal{S}_2}. \quad (54)$$

The constants can be chosen with $\delta < 1$, so $I + K$ is automatically invertible. Under the weaker assumptions (50), the same conclusion holds for orthogonal $I + K$ whenever f is non-Gaussian. This includes finite-rank perturbations whose ranges have infinite coordinate support.

Proof. In the general linear case, take $K_n = P_n K P_n$ and apply (35) to its finite coordinate block. Its untouched tail factors contribute affinity one, so the finite-dimensional Hellinger distance is also $h(\nu_{I+K_n}, \mu)$. The likelihood-ratio convergence in proposition 3.4 gives

$$h(\nu_{I+K_n}, \mu) = \left\| \sqrt{\frac{d\nu_{I+K_n}}{d\mu}} - 1 \right\|_2 \longrightarrow \left\| \sqrt{\frac{d\nu_{I+K}}{d\mu}} - 1 \right\|_2 = h(\nu_{I+K}, \mu).$$

Also $\|K_n\|_{\mathcal{S}_2} \rightarrow \|K\|_{\mathcal{S}_2}$. Passing to the limit proves both bounds.

In the orthogonal case, use the finite-coordinate polar approximants V_n and their likelihood-ratio convergence from proposition 3.5. They satisfy $V_n \rightarrow I + K$ in Hilbert–Schmidt norm. Choose the radius in this assertion to be half the finite-dimensional radius in (52). Then all sufficiently large V_n lie in that Frobenius ball, and

$$h(\nu_{V_n}, \mu) \longrightarrow h(\nu_{I+K}, \mu), \quad \|V_n - I\|_{\mathcal{S}_2} \longrightarrow \|K\|_{\mathcal{S}_2}.$$

Pass to the limit in (52). □

3.5.1 Characteristic-function rigidity and bounded invertible linear mixing

Necessity begins by identifying what a single output row can look like if its law is close to the original coordinate law. This is a one-dimensional rigidity problem for linear combinations of independent copies of X . The characteristic-function argument below says that, for a non-Gaussian coordinate law, such a row must asymptotically be one signed coordinate.

Throughout this subsection, X_j are independent copies of a real random variable X with law ρ and

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1,$$

and ρ is non-Gaussian. The characteristic-function arguments below require no moments of order greater than two. Recall

$$\phi(t) = \mathbb{E}e^{itX}, \quad \mu = \rho^{\otimes \mathbb{N}}.$$

The group $\mathcal{P}$ consists of coordinate permutations whose row signs preserve ρ . This definition applies even when ρ has no density.

Lemma 3.7 (A product approximation for small coefficients). *There is a nonnegative function ω_ρ , with $\omega_\rho(\delta) \rightarrow 0$ as $\delta \downarrow 0$, such that, for every finite or countable real family (b_j) with $\sum_j b_j^2 < \infty$,*

$$\left| \prod_j \phi(tb_j) - \exp\left(-\frac{t^2}{2} \sum_j b_j^2\right) \right| \leq \omega_\rho(\delta) t^2 \sum_j b_j^2 \quad \text{if} \quad \sup_j |tb_j| \leq \delta. \quad (55)$$

For a countable family, the product denotes the characteristic function of the L^2 sum $\sum_j b_j X_j$.

Proof. The centered second-moment assumption gives

$$\phi(u) = 1 - \frac{u^2}{2} + o(u^2).$$

Consequently

$$\omega_\rho(\delta) := \sup_{0 < |u| \leq \delta} \frac{|\phi(u) - e^{-u^2/2}|}{u^2} \longrightarrow 0, \quad \omega_\rho(0) := 0.$$

Both $\phi(u)$ and $e^{-u^2/2}$ have modulus at most one. Telescoping the difference of their finite products therefore bounds it by the sum of the differences of the factors, proving (55) for a finite family. For a countable family, pass to the limit through finite partial sums in L^2 and hence in distribution. This argument does not take a logarithm of the characteristic function. $\square$

Lemma 3.8 (Qualitative rigidity of a unit row). *Let $a^{(n)} \in \ell^2$ satisfy $\|a^{(n)}\|_2 = 1$. If*

$$\text{Law}\left(\sum_j a_j^{(n)} X_j\right) \Longrightarrow \rho,$$

then

$$\inf_{\substack{j \geq 1, \zeta \in \{-1, 1\} \\ \text{Law}(\zeta X) = \rho}} \|a^{(n)} - \zeta e_j\|_2 \longrightarrow 0. \quad (56)$$

Proof. We first describe the limits of such coefficient arrays. Permuting coefficients does not change the law of their sum. Arrange each row in nonincreasing order of absolute value, retaining its signs and padding with zeros if necessary. Its j th coefficient then has modulus at most $j^{-1/2}$. From any subsequence we may extract a further subsequence on which each coefficient converges, say $a_j^{(n)} \rightarrow a_j$. Fatou's lemma gives

$$\tau_a := \sum_j a_j^2 \leq 1.$$

For a fixed m , apply lemma 3.7 to the coefficients with $j > m$. For every fixed t , the error in replacing their characteristic function by

$$\exp\left[-\frac{t^2}{2} \left(1 - \sum_{j \leq m} (a_j^{(n)})^2\right)\right]$$

is at most $t^2\omega_\rho(|t|/\sqrt{m+1})$, uniformly in n . Fix any t . First let $n \rightarrow \infty$, and then let $m \rightarrow \infty$. The limiting characteristic function is

$$e^{-(1-\tau_a)t^2/2} \prod_j \phi(a_j t). \quad (57)$$

Equivalently, the limiting law is that of $\sum_j a_j X_j$ plus an independent centered Gaussian of variance $1 - \tau_a$.

Under the convergence hypothesis this limiting law is the law of X . Suppose that no $|a_j|$ is one. If all a_j vanish, (57) already makes X Gaussian. Otherwise the maximum $m_a := \max_j |a_j|$ exists, since $(a_j) \in \ell^2$, and satisfies $0 < m_a < 1$. For $\mathbf{j} = (j_1, \dots, j_k) \in \mathbb{N}^k$, set $b_{\mathbf{j}} := a_{j_1} \cdots a_{j_k}$. Iterating the characteristic-function identity (57) k times gives

$$\phi(t) = e^{-(1-\tau_a^k)t^2/2} \prod_{\mathbf{j} \in \mathbb{N}^k} \phi(tb_{\mathbf{j}}), \quad (58)$$

Here

$$\sum_{\mathbf{j}} b_{\mathbf{j}}^2 = \tau_a^k, \quad \sup_{\mathbf{j}} |b_{\mathbf{j}}| \leq m_a^k.$$

The identity may be obtained by substituting independent copies into the L^2 series; at each finite depth all series have summable coefficient squares. Applying lemma 3.7 to the last product in (58) yields

$$|\phi(t) - e^{-t^2/2}| \leq t^2 \tau_a^k \omega_\rho(|t| m_a^k) \rightarrow 0.$$

This again makes X Gaussian. This is a contradiction. Thus some $a_j = \varsigma \in \{-1, 1\}$; all the remaining coefficients and the independent Gaussian summand vanish. The limiting identity then says $\text{Law}(\varsigma X) = \rho$, so the sign preserves ρ .

To conclude, suppose (56) failed. Take a subsequence whose distance from all the indicated unit vectors is bounded below. The preceding extraction produces a coefficient converging to an allowed sign of modulus one. Since the row norms are one, its squared distance from that signed unit vector tends to zero:

$$\|a^{(n)} - \varsigma e_j\|_2^2 = 2 - 2\varsigma a_j^{(n)} \rightarrow 0.$$

Undoing the coefficient permutations gives the required contradiction. $\square$

Lemma 3.9 (Uniform invariance under tail symmetries). *Let μ be the i.i.d. product law and let $\nu \ll \mu$. Let $\mathcal{T}_N$ consist of the members of $\mathcal{P}$ that change only finitely many coordinates and fix coordinates $1, \dots, N$. Then*

$$\sup_{S \in \mathcal{T}_N} h(\nu, S_{\#}\nu) \rightarrow 0. \quad (59)$$

Also, the one-coordinate marginals of ν converge to the common coordinate law of μ , uniformly over coordinates greater than N .

Proof. Put $\ell = d\nu/d\mu$ and $\ell_N = \mathbb{E}_\mu[\ell \mid \sigma(x_1, \dots, x_N)]$. The martingale convergence theorem gives $\|\ell - \ell_N\|_{L^1(\mu)} \rightarrow 0$. Every $S \in \mathcal{T}_N$ preserves μ and satisfies $\ell_N \circ S^{-1} = \ell_N$. Thus

$$h(\nu, S_{\#}\nu)^2 \leq \|\ell - \ell \circ S^{-1}\|_{L^1(\mu)} \leq 2\|\ell - \ell_N\|_{L^1(\mu)},$$

which proves (59). For $i > N$, the marginal density is $\mathbb{E}_\mu[\ell \mid x_i]$. Independence and $\mathbb{E}_\mu \ell_N = 1$ imply $\mathbb{E}_\mu[\ell_N \mid x_i] = 1$, so

$$\|\mathbb{E}_\mu[\ell \mid x_i] - 1\|_{L^1(\mu)} \leq \|\ell - \ell_N\|_{L^1(\mu)}.$$

This proves the uniform marginal assertion. $\square$

Lemma 3.10 (Compactness of the covariance defect). *Let T be bounded on ℓ^2 . If $\nu_T \ll \mu$, then*

$$TT^* - I \text{ is compact.}$$

*If T is boundedly invertible, $T^*T - I$ is compact as well.*

Proof. We first record uniform integrability of the squares of the linear sums

$$Y_b := \sum_j b_j X_j, \quad \|b\|_2 \leq B, \quad (60)$$

for each fixed $B < \infty$. Let

$$X^{(M)} = X \mathbf{1}_{\{|X| \leq M\}} - \mathbb{E}[X \mathbf{1}_{\{|X| \leq M\}}], \quad Y_b^{(M)} = \sum_j b_j X_j^{(M)}.$$

The variables $X^{(M)}$ are bounded and centered, and converge to X in L^2 . Independence gives

$$\sup_{\|b\|_2 \leq B} \|Y_b - Y_b^{(M)}\|_2 \leq B \|X - X^{(M)}\|_2 \rightarrow 0.$$

For fixed M , expanding the fourth moment of a sum of independent centered bounded variables gives

$$\sup_{\|b\|_2 \leq B} \mathbb{E}|Y_b^{(M)}|^4 < \infty.$$

The bound first holds for finite sums; it also makes their truncations Cauchy in L^4 , proving the countable version. Moreover,

$$\sup_{\|b\|_2 \leq B} \|Y_b^2 - (Y_b^{(M)})^2\|_1 \leq B^2 \|X - X^{(M)}\|_2 (\|X\|_2 + \|X^{(M)}\|_2) \rightarrow 0.$$

Thus the squares in (60) are uniformly approximated in L^1 by uniformly integrable families, and are themselves uniformly integrable. This uses fourth moments of bounded truncations, not a fourth-moment assumption on X .

Write $\nu = \nu_T$, $\ell = d\nu/d\mu$, and $\ell_N = \mathbb{E}_\mu[\ell \mid \sigma(x_1, \dots, x_N)]$. For a finitely supported unit vector a supported on coordinates greater than N , put $F_a(x) = (\sum_i a_i x_i)^2$. Under μ , these are squares from (60) with $B = 1$; under ν , they are squares of Y_{T^*a} under μ , with $\|T^*a\|_2 \leq \|T\|_{\text{op}}$. They are therefore uniformly integrable under both laws. The truncated statistic $F_a \wedge L$ depends only on tail coordinates. Independence yields

$$|\mathbb{E}_\nu(F_a \wedge L) - \mathbb{E}_\mu(F_a \wedge L)| \leq L \|\ell - \ell_N\|_1.$$

First let $N \rightarrow \infty$ with fixed L . Then let $L \rightarrow \infty$. Uniform integrability gives

$$\sup_{\|a\|_2=1, \text{ } a \text{ finitely supported}} |\langle (TT^* - I)a, a \rangle| \rightarrow 0.$$

Since $TT^* - I$ is self-adjoint and bounded, density of finite vectors in the tail subspace implies

$$\|(I - P_N)(TT^* - I)(I - P_N)\|_{\text{op}} \rightarrow 0.$$

The complementary rows and columns form a finite-rank operator. Hence $TT^* - I$ is a norm limit of finite-rank operators and is compact. For invertible T , use $T^*T - I = T^{-1}(TT^* - I)T$. $\square$

We now assemble the necessity argument. The preceding row rigidity gives only qualitative matching of far-out rows with signed coordinates. To obtain a Hilbert–Schmidt conclusion, those row errors must be summable. The proof forces summability by swapping carefully chosen tail coordinates: if the errors were not summable, the swaps would create a small finite-rank perturbation that local coercivity can detect, while tail invariance makes it asymptotically invisible.

Lemma 3.11 (Hilbert–Schmidt necessity under a second-moment hypothesis). *Assume that the common law has an a.e.-positive density f , that $g = \sqrt{f} \in W^{1,2}(\mathbb{R})$, and that $xg' \in L^2(\mathbb{R})$. If $T \in \text{GL}(H)$ and $\nu_T \sim \mu$, then*

$$T - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}. \quad (61)$$

When T is orthogonal, the condition $xg' \in L^2$ can be omitted.

Proof. By lemma 3.10, $\|T^*e_i\|_2 \rightarrow 1$. By lemma 3.9, the law of $\sum_j (T^*e_i)_j X_j$ converges to ρ . Normalizing T^*e_i to a unit vector changes this sum by a quantity tending to zero in L^2 . Therefore lemma 3.8 gives indices $\iota(i)$ and signs $\varsigma_i \in \{-1, 1\}$ preserving ρ , outside a finite initial set, with

$$\widehat{e}_i := \varsigma_i e_{\iota(i)}, \quad \|T^*e_i - \widehat{e}_i\|_2^2 \rightarrow 0. \quad (62)$$

The indices $\iota(i)$ are distinct sufficiently far out. Indeed, if $\iota(i) = \iota(j)$ for two such rows, then $\varsigma_i T^*e_i - \varsigma_j T^*e_j$ has arbitrarily small norm by (62), whereas bounded invertibility gives the lower bound $\sqrt{2}/\|T^{-1}\|_{\text{op}}$. Hence we can fix N_0 so that ι is injective on $i > N_0$.

There is also column approximation along this matching. Compactness of $T^*T - I$ and the fact that $\iota(i)$ leaves every finite set imply

$$\|T\widehat{e}_i\|_2^2 \rightarrow 1.$$

Since $\langle T\widehat{e}_i, e_i \rangle = \langle \widehat{e}_i, T^*e_i \rangle \rightarrow 1$, we obtain $\|T\widehat{e}_i - e_i\|_2^2 \rightarrow 0$. Consequently

$$\text{err}_i := \|T^*e_i - \widehat{e}_i\|_2^2 + \|T\widehat{e}_i - e_i\|_2^2 \rightarrow 0. \quad (63)$$

We prove that $\sum_i \text{err}_i < \infty$ on this tail. By lemma 3.6, there are $c_{\text{HS}}, \delta_{\text{HS}} > 0$ such that

$$h(\mu, \nu_W) \geq c_{\text{HS}} \|W - I\|_{\mathcal{S}_2} \quad \text{when } W \in \text{GL}(H), \quad \|W - I\|_{\mathcal{S}_2} \leq \delta_{\text{HS}}. \quad (64)$$

Suppose instead that $\sum_i \text{err}_i = \infty$. Choose $\delta > 0$ small enough that $3\|T^{-1}\|_{\text{op}}\delta < \delta_{\text{HS}}$. In any sufficiently remote tail, where $\text{err}_i \leq \delta^2$, choose a finite set E satisfying

$$\delta^2 \leq \sum_{i \in E} \text{err}_i \leq 2\delta^2. \quad (65)$$

Pair each $i \in E$ with a distinct auxiliary index k_i beyond E , and set $G = \{k_i : i \in E\}$. Since $\text{err}_i \rightarrow 0$, these auxiliary indices may be chosen with an arbitrarily small value of $\sum_{k \in G} \text{err}_k$, while E remains fixed.

Let S exchange each output pair (e_i, e_{k_i}) , and let P exchange the aligned signed input pairs $(\widehat{e}_i, \widehat{e}_{k_i})$, acting identically on their orthogonal complements. Both are involutions in $\mathcal{P}$, so they preserve μ . Define

$$W = PT^{-1}ST.$$

Then

$$W - I = PT^{-1}(ST - TP), \quad \frac{\|ST - TP\|_{\mathcal{S}_2}}{\|T\|_{\text{op}}} \leq \|W - I\|_{\mathcal{S}_2} \leq \|T^{-1}\|_{\text{op}} \|ST - TP\|_{\mathcal{S}_2}. \quad (66)$$

These are finite-rank differences. Indeed $S - I$ and $P - I$ have finite rank. Hellinger invariance and $P_{\#}\mu = \mu$ give

$$h(\mu, \nu_W) = h(\nu_T, S_{\#}\nu_T). \quad (67)$$

All maps in use lie in a countable operator group, so [lemma 3.1](#) justifies this identity on a common carrier.

We now estimate $ST - TP$ in (66). Obtain T_0 from T by replacing each auxiliary row and column with its matched coordinate vector:

$$T_0^* e_k = \widehat{e}_k, \quad T_0 \widehat{e}_k = e_k \quad (k \in G).$$

On the orthogonal complements of these input and output coordinates, retain the corresponding block of T . These conditions uniquely determine T_0 . Orthogonal block decomposition then gives

$$\|T - T_0\|_{\mathcal{S}_2}^2 \leq \sum_{k \in G} (\|T^* e_k - \widehat{e}_k\|_2^2 + \|T \widehat{e}_k - e_k\|_2^2) = \sum_{k \in G} \text{err}_k. \quad (68)$$

For completeness, the left side counts the squared entries of the auxiliary row and column errors once, while their intersection is counted twice on the right.

Order the output subspaces as E , its complement outside $E \cup G$, and G . Order the input subspaces as the aligned signed coordinates for E , their complement outside the aligned $E \cup G$, and the aligned coordinates for G . In these orthonormal decompositions,

$$T_0 = \begin{pmatrix} A & B & 0 \\ C & D & 0 \\ 0 & 0 & I \end{pmatrix}, \quad ST_0 - T_0 P = \begin{pmatrix} 0 & -B & I - A \\ C & 0 & -C \\ A - I & B & 0 \end{pmatrix}.$$

The blocks B, C have finite-dimensional range or domain and are Hilbert–Schmidt. The sum of squared row and column errors on E is

$$\sum_{i \in E} (\|T_0^* e_i - \widehat{e}_i\|_2^2 + \|T_0 \widehat{e}_i - e_i\|_2^2) = 2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2.$$

The displayed block matrix gives the exact identity

$$\|ST_0 - T_0 P\|_{\mathcal{S}_2}^2 = 2(\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2),$$

and therefore

$$\begin{aligned} 2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2 &\leq \|ST_0 - T_0 P\|_{\mathcal{S}_2}^2 \\ &\leq 2(2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2). \end{aligned} \quad (69)$$

As the auxiliary error in (68) tends to zero, the displayed sum of squared errors tends to $\sum_{i \in E} \text{err}_i$. Also,

$$\|(ST - TP) - (ST_0 - T_0 P)\|_{\mathcal{S}_2} \leq 2\|T - T_0\|_{\mathcal{S}_2} \longrightarrow 0.$$

By (65) and (69), we can choose actual finite auxiliary indices so that

$$\frac{\delta}{2} \leq \|ST - TP\|_{\mathcal{S}_2} \leq 3\delta.$$

It follows from (66) that

$$\frac{\delta}{2\|T\|_{\text{op}}} \leq \|W - I\|_{\mathcal{S}_2} \leq 3\|T^{-1}\|_{\text{op}}\delta < \delta_{\text{HS}}.$$

The local bound (64) now forces $h(\mu, \nu_W) \geq c_{\text{HS}}\delta/(2\|T\|_{\text{op}})$. On the other hand, S is supported in the original remote tail. lemma 3.9 makes the right side of (67) uniformly smaller than this positive number when that tail is far enough out. This is the contradiction. Therefore $\sum_i \text{err}_i < \infty$.

Define Q_0 by the matched rows $\hat{e}_i$ after the fixed initial cutoff N_0 , and by zero rows for $i \leq N_0$. The summability just proved implies

$$\begin{aligned} \|T - Q_0\|_{\mathcal{S}_2}^2 &= \sum_{i \leq N_0} \|T^*e_i\|_2^2 + \sum_{i > N_0} \|T^*e_i - \hat{e}_i\|_2^2 \\ &\leq \sum_{i \leq N_0} \|T^*e_i\|_2^2 + \sum_{i > N_0} \text{err}_i < \infty. \end{aligned}$$

Therefore Q_0 is a compact perturbation of an invertible operator and is Fredholm of index zero. Its cokernel consists precisely of the first N_0 output coordinates. Its kernel is spanned by the unused input coordinates. There are exactly N_0 unused columns. Match them to the first N_0 rows with sign $+1$. This completes Q_0 to a single $Q \in \mathcal{P}$ and proves (61).

If T is orthogonal, both covariance defects vanish, and every operator $W = PT^*ST$ constructed above is orthogonal. Consequently only the orthogonal part of lemma 3.6 is needed. That part uses location Fisher information alone, so the condition $xg' \in L^2$ may be omitted in this case. $\square$

3.6 Exact identification of the linear stabilizer

Unlike measure-class equivalence, exact invariance requires only the second moment.

Proof of corollary 1.3. Every member of $\mathcal{P}$ preserves the product law. Conversely, if $\nu_T = \mu$, covariance gives $TT^* = I$. Since T is boundedly invertible, it is orthogonal. Each of its unit rows gives a linear combination with the same law as X . Applying lemma 3.8 to the constant sequence consisting of this row shows that the row is an allowed signed unit vector. Orthogonality makes the selected coordinates a bijection. Thus $T \in \mathcal{P}$. $\square$

3.7 Two-sided uniform integrability and completion of the proof

The finite-coordinate limits and the two finite-marginal conditional-expectation martingales now combine to complete the proof.

Proof of theorem 1.1 and proposition 1.2. Necessity follows from lemma 3.11. If $T - Q \in \mathcal{S}_2$ for $Q \in \mathcal{P}$, then $TQ^{-1} = I + (T - Q)Q^{-1}$ is invertible and differs from the identity by a Hilbert–Schmidt operator. By proposition 3.4, $\nu_{TQ^{-1}} \sim \mu$. Since $Q_{\#}\mu = \mu$, this is $\nu_T \sim \mu$. The same proposition gives the finite-coordinate likelihood-ratio and square-root limits

stated in [proposition 1.2](#). The exact linear stabilizer is identified in [corollary 1.3](#). Finally, [lemmas 3.2](#) and [3.3](#) give the finite-marginal affinity characterization and the singular alternative in [theorem 1.1](#), and both uniform-integrability statements in [proposition 1.2](#). $\square$

Proof of [corollary 1.4](#). Necessity is [lemma 3.11](#). For sufficiency, write $UQ^{-1} = I + (U - Q)Q^{-1}$. This is orthogonal and differs from the identity by a Hilbert–Schmidt operator. Apply [proposition 3.5](#), and then use $Q_{\#}\mu = \mu$. The alternative follows from [lemma 3.2](#). $\square$

3.8 Globalization of finite-dimensional coercivity

The infinite-dimensional classification supplies the compactness needed to rule out finite-dimensional almost-symmetries that stay a fixed distance from the exact linear stabilizer.

Proof of [theorem 2.7](#). Fix $0 < a \leq b < \infty$. We first prove uniform separation away from the exact linear stabilizer: for every $\varepsilon > 0$,

$$\inf \left\{ h(\mu_n, A_{\#}\mu_n)^2 : \begin{array}{l} n \geq 1, A \in \text{GL}(n, \mathbb{R}), \\ s_{\min}(A) \geq a, s_{\max}(A) \leq b, \\ d_n(A) \geq \varepsilon \end{array} \right\} > 0. \quad (70)$$

Suppose otherwise. Then there are dimensions n_k and matrices A_k in the displayed class with $h(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) \rightarrow 0$. Since

$$-2 \log A(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) = -2 \log \left(1 - \frac{1}{2} h(\mu_{n_k}, (A_k)_{\#}\mu_{n_k})^2 \right),$$

we may pass to a subsequence. Without changing notation, assume this sum of block energies is finite.

Decompose $H = \bigoplus_{k \geq 1} \mathbb{R}^{n_k}$ into consecutive coordinate blocks. Now define $A = \bigoplus_{k \geq 1} A_k$. The common singular-value bounds imply $A \in \text{GL}(H)$. Each block law $(A_k)_{\#}\mu_{n_k}$ is equivalent to μ_{n_k} because $f > 0$ almost everywhere and A_k is invertible. The summability of the block energies gives

$$\prod_{k \geq 1} A(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) > 0.$$

Kakutani’s product theorem [\[10\]](#) therefore gives $\nu_A \sim \mu$. The already proved [theorem 1.1](#) therefore supplies $Q \in \mathcal{P}$ such that

$$D := A - Q \in \mathcal{S}_2(H). \quad (71)$$

Let Π_k be the orthogonal projection onto the k -th output block. The orthogonality of these blocks gives

$$\sum_k \|\Pi_k D\|_{\mathcal{S}_2}^2 = \|D\|_{\mathcal{S}_2}^2 < \infty, \quad \text{hence} \quad \|\Pi_k D\|_{\mathcal{S}_2} \rightarrow 0.$$

If, for some output basis vector e_i in the k -th block, the signed basis vector Q^*e_i lies outside the corresponding input block, then A^*e_i and Q^*e_i have disjoint supports. The lower singular value bound then yields

$$\|A^*e_i - Q^*e_i\|_2^2 = \|A_k^*e_i\|_2^2 + 1 \geq a^2 + 1. \quad (72)$$

For all sufficiently large k , this contradicts $\|\Pi_k D\|_{\mathcal{S}_2}^2 < a^2 + 1$. Hence Q^* maps every output coordinate of each sufficiently large block into the same input block. Its restriction is injective

between two sets of cardinality n_k . Hence it is a bijection. Denote the resulting member of $\mathcal{P}_{n_k}$ by Q_k . It follows that

$$d_{n_k}(A_k) \leq \|A_k - Q_k\|_F = \|\Pi_k D\|_{\mathcal{S}_2} \longrightarrow 0,$$

contradicting $d_{n_k}(A_k) \geq \varepsilon$. This proves (70).

Let $c_{\text{loc}}, C_{\text{loc}}, \delta_{\text{loc}}$ be the constants in [proposition 2.6](#), decreasing δ_{loc} so that $0 < \delta_{\text{loc}} < 1$. Given T , choose a minimizer $Q_n \in \mathcal{P}_n$ in (36). Since $(Q_n)_{\#}\mu_n = \mu_n$,

$$h(\mu_n, T_{\#}\mu_n) = h(\mu_n, (TQ_n^{-1})_{\#}\mu_n), \quad \|TQ_n^{-1} - I\|_F = d_n(T).$$

When $d_n(T) \leq \delta_{\text{loc}}$, the local theorem supplies the two-sided linear bounds for h . When $d_n(T) > \delta_{\text{loc}}$, (70) with $\varepsilon = \delta_{\text{loc}}$ supplies the lower bound, while $h^2 \leq 2$ supplies the upper bound. Choose $c > 0$ no larger than c_{loc}^2 and the positive infimum in (70) with $\varepsilon = \delta_{\text{loc}}$, and take

$$C = \max\{C_{\text{loc}}^2, 2\delta_{\text{loc}}^{-2}\}.$$

These constants give (37) in all regimes. □

4 Symmetric Stable Products and Anisotropic Geometry

The finite-variance criterion does not extend unchanged to symmetric stable products: both the coefficient domain and the mixing cost become α -dependent. Fix $0 < \alpha < 2$, let ρ_α be the standard symmetric α -stable law, and put

$$\int_{\mathbb{R}} e^{itx} \rho_\alpha(dx) = e^{-|t|^\alpha}, \quad \mu_\alpha = \rho_\alpha^{\otimes \mathbb{N}}, \quad \mu_{\alpha,n} = \rho_\alpha^{\otimes n}. \quad (73)$$

Let $(X_j)_{j \geq 1}$ be independent with common law ρ_α , and write $\mathbf{X} = (X_j)_{j \geq 1}$. Its density f_α is smooth and strictly positive, and its location and scale Fisher information are finite, although its second moment is infinite [1]. Indeed, $f_\alpha(x) \asymp (1+|x|)^{-1-\alpha}$ and its location score is $O((1+|x|)^{-1})$ in the tails, so both the location score and its product with x are square integrable against f_α . For $\alpha \leq 1$, the normalization in (73) specifies a zero location parameter, not a finite mean.

A stable row series $\sum_j a_j X_j$ converges almost surely precisely when $a \in \ell^\alpha$. For a real matrix T with ℓ^α rows, write $\Phi_{\alpha,T}$ for the associated row-series map and $\nu_{\alpha,T} = (\Phi_{\alpha,T})_{\#}\mu_\alpha$. We say that T admits a compatible stable row-series realization if there is a matrix S with ℓ^α rows such that the two row-series maps are measurable and

$$\Phi_{\alpha,S} \circ \Phi_{\alpha,T} = I, \quad \Phi_{\alpha,T} \circ \Phi_{\alpha,S} = I$$

on common conull carriers for the corresponding orbit laws. This is the stable analogue of the compatible realization supplied by [lemma 3.1](#); it is an assumption on the row-series realization, not a claim that every bounded ℓ^2 operator has stable row sums.

For a matrix K , define its column cost by

$$F_\alpha(K) = \sum_j \|Ke_j\|_2^\alpha.$$

For $M = (m_{ij})$, define the anisotropic mixing cost

$$\mathfrak{e}_\alpha(M) = \sum_i |m_{ii} - 1|^2 + F_\alpha(M - \text{diag } M). \quad (74)$$

The first term measures coordinate scales. The second takes the ℓ^2 -norm down each off-diagonal input column and then its α -th power across source coordinates. For the actual stable output marginals, set

$$E_{\alpha,n}(T) = -2 \log A(\mu_{\alpha,n}, (\pi_n)_\# \nu_{\alpha,T}). \quad (75)$$

Theorem 4.1 (Complete criterion for symmetric stable coordinates). *Let $0 < \alpha < 2$ and suppose that T admits a compatible stable row-series realization. Then*

$$\boxed{\begin{aligned} \nu_{\alpha,T} \sim \mu_\alpha &\iff T|_H \in \text{GL}(H) \text{ and } \mathfrak{e}_\alpha(TQ^{-1}) < \infty \text{ for some } Q \in \mathcal{P}^\pm \\ &\iff \sup_n E_{\alpha,n}(T) < \infty, \end{aligned}} \quad (76)$$

Here $\mathfrak{e}_\alpha$ is the anisotropic mixing cost from (74), whose off-diagonal component is the column cost F_α . If these conditions fail, then $\nu_{\alpha,T} \perp \mu_\alpha$ and $E_{\alpha,n}(T) \uparrow \infty$. Conversely, if $T \in \text{GL}(H)$ satisfies the summability condition in (76), that condition itself supplies the compatible stable row-series realization for T and its inverse. The exact linear stabilizer in this class is $\mathcal{P}^\pm$.

Proof. We organize the proof around the four points at which the finite-variance argument must be replaced.

Dichotomy, energy, and the coefficient domain. Put $g_\alpha = \sqrt{f_\alpha}$ and define

$$a_\alpha(u) = \int_{\mathbb{R}} g_\alpha(x) g_\alpha(x - u) dx.$$

Finite positive location Fisher information, everywhere positivity of the density, and decay of translations in L^2 give

$$1 - a_\alpha(u) \asymp_\alpha \min\{u^2, 1\}. \quad (77)$$

Kakutani's theorem therefore identifies ℓ^2 as exactly the space of translations under which μ_α is quasi-invariant. If $\nu_{\alpha,T} \sim \mu_\alpha$, translating before and after the compatible stable row-series realization shows that T and its row inverse map this translation space into itself. The closed graph theorem then gives $T|_H \in \text{GL}(H)$.

The compatible two-sided inverse makes every finite family of output rows linearly independent. It therefore has a nonsingular input minor, and the remaining stable inputs contribute an independent convolution factor; hence every finite output block has an everywhere-positive density. The finite-marginal likelihood ratios consequently form the same mean-one martingale as in lemma 3.3. The product measure is quasi-invariant and ergodic under finitely supported translations, while its row-series image has the same properties under their images by T . The equivalent-singular theorem for quasi-invariant ergodic measures allows these two translation spaces to differ and therefore applies [11]. It follows exactly as in (42) that

$$\nu_{\alpha,T} \sim \mu_\alpha \iff \sup_n E_{\alpha,n}(T) < \infty, \quad (78)$$

with singularity and divergent energy on the other branch.

Sufficiency. The column cost satisfies

$$F_\alpha(AK) \leq \|A\|_{\text{op}}^\alpha F_\alpha(K), \quad F_\alpha(KD) \leq \|D\|_{\text{op}}^\alpha F_\alpha(K) \quad (79)$$

for bounded A and bounded diagonal D , and

$$\|K^* \gamma\|_\alpha^\alpha \leq F_\alpha(K) \|\gamma\|_2^\alpha. \quad (80)$$

Thus $F_\alpha(K) < \infty$ implies that K has admissible stable rows.

The finite-dimensional estimate that replaces the matrix-Fisher upper bound is

$$h^2(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}) \leq C_\alpha \min\{\mathfrak{e}_\alpha(A), 1\} \quad (81)$$

for every invertible real $n \times n$ matrix A , with a constant independent of n . To prove it, put $\beta = \alpha/2$, let V_i be independent positive β -stable variables with $\mathbb{E}e^{-tV_i} = e^{-t^\beta}$, and write $X_i = \sqrt{2V_i}Z_i$ with independent standard Gaussians Z_i . Conditionally on the diagonal matrix V with entries V_i , Hellinger data processing and the Gaussian affinity formula bound the left side of (81) by a constant times

$$\mathbb{E} \min\{\|V^{-1/2}(A - I)V^{1/2}\|_F^2, 1\}.$$

For $c_j^2 = \sum_{i \neq j} |a_{ij}|^2$, the contribution of source column j is

$$W_j = V_j \sum_{i \neq j} \frac{|a_{ij}|^2}{V_i}.$$

The positive stable estimates, for $y > 0$,

$$\mathbb{P}(V_1 > y) \leq C_\beta y^{-\beta}, \quad \mathbb{E}[V_1; V_1 \leq y] \leq C_\beta y^{1-\beta}, \quad \mathbb{E}V_1^{-1} < \infty$$

give

$$\mathbb{P}(c_j^2 V_j > 1) + \mathbb{E}[W_j; c_j^2 V_j \leq 1] \leq C_\alpha c_j^\alpha.$$

Summing over source columns, while retaining the ordinary quadratic bound on the diagonal, proves (81).

Now remove the measure-preserving Q and write the resulting matrix as $M = D + K$, where D is diagonal with entries d_i , $(d_i - 1) \in \ell^2$, and $F_\alpha(K) < \infty$. The finitely many indices on which the diagonal of M is not positive are absorbed into K . For

$$D_n = I + P_n(D - I)P_n, \quad K_n = P_n K P_n, \quad M_n = D_n + K_n,$$

we have $M_n \rightarrow M$ in operator norm, so the M_n are eventually invertible with uniformly bounded inverses. The identity

$$M_n^{-1} = D_n^{-1} - M_n^{-1} K_n D_n^{-1}$$

and the column-class bounds show that $F_\alpha(M_n^{-1} K_n D_n^{-1})$ is uniformly bounded. Moreover,

$$\begin{aligned} M_n^{-1} M_m - I &= D_n^{-1}(D_m - D_n) - M_n^{-1} K_n D_n^{-1}(D_m - D_n) \\ &\quad + M_n^{-1}(K_m - K_n). \end{aligned} \quad (82)$$

The first term has diagonal ℓ^2 -norm tending to zero. The other two have column cost tending to zero by (79), $\|D_m - D_n\|_{\text{op}} \rightarrow 0$, and $F_\alpha(K_m - K_n) \rightarrow 0$; their diagonal square cost tends to zero as well because finite column cost implies membership in $\mathcal{S}_2$. Thus (81), applied to $M_n^{-1} M_m$, makes the square-root likelihoods of $(M_n)_{\#}\mu_\alpha$ Cauchy in $L^2(\mu_\alpha)$. Their limit has norm one, and convergence of each stable row in ℓ^α identifies its square with $d\nu_{\alpha,M}/d\mu_\alpha$. This proves absolute continuity and then equivalence by (78). Finally,

$$(D + K)^{-1} = D^{-1} - (D + K)^{-1} K D^{-1} \quad (83)$$

and (79) show that the inverse has the same admissible form. The continuity of the stable random functional from ℓ^α to convergence in probability lets the finite-matrix composition identities pass to the row-series limits, producing compatible measurable versions. This also proves the last assertion of the theorem.

Reduction to a Hilbert–Schmidt perturbation. Suppose conversely that $\nu_{\alpha,T} \sim \mu_\alpha$. We transfer the problem to a finite-variance product without applying the finite-variance theorem directly to the stable law. Let $\widehat{g}_\alpha$ be the unitary Fourier transform of g_α , and set

$$\lambda(d\xi) = |\widehat{g}_\alpha(\xi)|^2 d\xi.$$

The probability λ is symmetric and has variance $\|g'_\alpha\|_2^2 \in (0, \infty)$. Moreover, $\widehat{g}_\alpha * \widehat{g}_\alpha$ is a positive constant multiple of $e^{-|t|^\alpha}$; hence $|\widehat{g}_\alpha|^2 * |\widehat{g}_\alpha|^2 > 0$ almost everywhere. The characteristic function of λ is a_α . Polynomial stable tails give, for some $m_0, \delta_0 > 0$,

$$c(1 + |t|)^{-m_0} \leq a_\alpha(t) \leq C(1 + |t|)^{-\delta_0},$$

and a'_α is bounded. Choose an integer r so large that

$$\int_{\mathbb{R}} t^2 (|a_\alpha(t)|^{2r} + |(a_\alpha^r)'(t)|^2) dt < \infty.$$

Fourier inversion makes the density of λ^{*r} a bounded-variation density; three such convolutions have finite location Fisher information by the Fisher smoothing estimate [1]. Let f_λ be the density of λ^{*3r} , put $f_{\lambda,2} = f_\lambda * f_\lambda$, and let $s_\lambda, s_{\lambda,2}$ be their location scores. Independent U_1, U_2 with density f_λ satisfy

$$xs_{\lambda,2}(x) - 1 = 2\mathbb{E}[U_1 s_\lambda(U_2) \mid U_1 + U_2 = x].$$

Their finite variance and the finite Fisher information of f_λ therefore give finite scale Fisher information for $f_{\lambda,2}$. Set $k = 6r$ and $\Lambda_k = (\lambda^{*k})^{\otimes \mathbb{N}}$. The density of λ^{*k} , after rescaling to variance one, is a.e. positive and satisfies (3). It is non-Gaussian because its characteristic function a_α^k has a polynomial lower bound, whereas a nondegenerate Gaussian characteristic function has Gaussian decay.

Let $f_{\alpha,n} = f_\alpha^{\otimes n}$, let $f_{\alpha,T,n}$ be the density of the true first n stable outputs, and put

$$\ell_{\alpha,T,n} = \frac{f_{\alpha,T,n}}{f_{\alpha,n}}, \quad \ell_{\alpha,T} = \frac{d\nu_{\alpha,T}}{d\mu_\alpha}.$$

Here $\ell_{\alpha,T,n}$ is lifted to the product space and $\sqrt{f_{\alpha,T,n}} = g_\alpha^{\otimes n} \sqrt{\ell_{\alpha,T,n}}$. Equivalence gives $\sqrt{\ell_{\alpha,T,n}} \rightarrow \sqrt{\ell_{\alpha,T}}$ in $L^2(\mu_\alpha)$.

For each n , take k tensor copies of $\sqrt{f_{\alpha,T,n}}$, apply the finite-dimensional unitary Fourier transform, square the modulus, and push the resulting probability on $\mathbb{R}^{kn}$ forward by adding the k frequency coordinates belonging to each site. Call the resulting probability on $\mathbb{R}^n$ $\xi_{\alpha,T,n}$, and extend it to the full sequence space by

$$\Xi_{\alpha,T,n} = \xi_{\alpha,T,n} \otimes (\lambda^{*k})^{\otimes \{i > n\}}. \quad (84)$$

The finite-dimensional pushforward $\xi_{\alpha,T,n}$ has a Lebesgue density, and the density of λ^{*k} is positive almost everywhere; hence $\Xi_{\alpha,T,n} \ll \Lambda_k$.

For $m > n$, pad $\sqrt{f_{\alpha,T,n}}$ by $m - n$ factors of g_α before making the Fourier comparison. Using total variation in the L^1 -norm convention, unitarity, the inequality $\| |u|^2 - |v|^2 \|_1 \leq 2\|u - v\|_2$ for unit vectors, and contraction of total variation under pushforward give

$$\|\Xi_{\alpha,T,m} - \Xi_{\alpha,T,n}\|_{\text{TV}} \leq 2\|(\sqrt{\ell_{\alpha,T,m}})^{\otimes k} - (\sqrt{\ell_{\alpha,T,n}})^{\otimes k}\|_{L^2(\mu_\alpha^{\otimes k})} \leq 2k\|\sqrt{\ell_{\alpha,T,m}} - \sqrt{\ell_{\alpha,T,n}}\|_{L^2(\mu_\alpha)}. \quad (85)$$

Thus $\Xi_{\alpha,T,n}$ converges in total variation to a probability $\Xi_{\alpha,T} \ll \Lambda_k$.

Let $\nu_{\alpha,T}^{[n]} = \ell_{\alpha,T,n}\mu_\alpha$; this measure has the true first n output marginal and the original independent tail. For every finitely supported z , Fourier autocorrelation and the square-root likelihood convergence give

$$\begin{aligned} \widehat{\Xi}_{\alpha,T}(z) &= \lim_n \mathbf{A}(\nu_{\alpha,T}^{[n]}, (x \mapsto x + z)_{\#} \nu_{\alpha,T}^{[n]})^k \\ &= \mathbf{A}(\nu_{\alpha,T}, (x \mapsto x + z)_{\#} \nu_{\alpha,T})^k \\ &= \prod_i a_\alpha((T^{-1}z)_i)^k. \end{aligned}$$

The last expression is the characteristic function of $((T^{-1})^*)_{\#} \Lambda_k$, so finite-dimensional uniqueness identifies $\Xi_{\alpha,T} = ((T^{-1})^*)_{\#} \Lambda_k$. Consequently

$$((T^{-1})^*)_{\#} \Lambda_k \ll \Lambda_k. \quad (86)$$

The equivalent–singular alternative in [theorem 1.1](#) upgrades this absolute continuity to equivalence. Applying that theorem to the product of the λ^{*k} law rescaled to variance one then gives $Q \in \mathcal{P}^\pm$ such that $(T^{-1})^* - Q \in \mathcal{S}_2$. Since

$$T - Q = -T((T^{-1})^* - Q)^*Q,$$

we obtain $T - Q \in \mathcal{S}_2$. After removing Q , write

$$TQ^{-1} = I + K, \quad K \in \mathcal{S}_2. \quad (87)$$

Necessity of the off-diagonal column summability. It remains to strengthen (87) from a quadratic conclusion to the second sum in (74). The local input is the following dimension-free estimate, valid for every dimension and every real square matrix B : there are $\varepsilon_\alpha, c_\alpha > 0$ such that

$$\text{diag } B = 0, \quad F_\alpha(B) \leq \varepsilon_\alpha \implies h^2((I + B)_{\#} \mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha F_\alpha(B). \quad (88)$$

To prove it, write $F = F_\alpha(B)$, $c_j = \|Be_j\|_2$, and $u_j = Be_j/c_j$ when $c_j \neq 0$. Then $\|u_j\|_2 = 1$, $(u_j)_j = 0$, and the stable scale contributed to output row i satisfies

$$\omega_i(B) := \sum_j |b_{ij}|^\alpha \leq F. \quad (89)$$

First, for any finite family of vectors v_j in a real Hilbert space,

$$\mathbb{P} \left(\left\| \sum_j v_j X_j \right\|_2 > t \right) \leq C_\alpha t^{-\alpha} \sum_j \|v_j\|_2^\alpha. \quad (90)$$

Indeed, exclude the events $\|v_j\|_2|X_j| > t$, truncate on their complement, and apply Chebyshev's inequality to the centered truncated sum. The stable tail and truncated second moment give (90). In particular, there is a deterministic modulus $\tau_\alpha(s) \downarrow 0$ as $s \downarrow 0$ such that

$$\mathbb{E} \min \left\{ \left\| \sum_j v_j X_j \right\|_2, 1 \right\} \leq \tau_\alpha \left(\sum_j \|v_j\|_2^\alpha \right). \quad (91)$$

Second, one bounded statistic detects a unit displacement in every Euclidean direction, uniformly in the dimension. Take $\psi(x) = \tanh x$. There are a smooth odd increasing function χ , bounded in modulus by one, and $c_{\alpha, \text{rec}} > 0$, depending only on ρ_α , such that for every unit vector u ,

$$\Theta_u(x) = \sum_i u_i \psi(x_i), \quad \mathcal{M}_u(t) = \mathbb{E} \chi(\Theta_u(\mathbf{X} + tu)), \quad \mathcal{M}_u(t) \geq c_{\alpha, \text{rec}} \quad (1 \leq t \leq 2). \quad (92)$$

To verify this, note that $\Theta_u(\mathbf{X})$ has mean zero and a variance independent of u . Choose a common interval carrying most of its mass and then choose χ whose derivative has a positive lower bound on the enlarged interval. Since

$$\inf_{|\zeta| \leq 2} \mathbb{E} \psi'(X + \zeta) > 0,$$

differentiation gives a dimension-free positive lower bound for

$$\mathcal{M}'_u(t) = \mathbb{E} \left[\chi'(\Theta_u(\mathbf{X} + tu)) \sum_i u_i^2 \psi'(X_i + tu_i) \right]$$

when $|t| \leq 2$, after discarding the common small exceptional event. Oddness then proves (92).

Third, choose a nonzero smooth odd function w , supported where $1 \leq |t| \leq 2$, with $0 \leq w \leq 1$ on the positive half-line. Put

$$\Theta_j(x) = \sum_{i \neq j} (u_j)_i \psi(x_i),$$

and define the bounded test

$$\Psi_B(x) = \max \left\{ -1, \min \left\{ \sum_{j: c_j \neq 0} w(c_j x_j) \chi(\Theta_j(x)), 1 \right\} \right\}. \quad (93)$$

For all sufficiently small c_j , stable tail asymptotics give

$$\mathbb{P}(1 \leq |c_j X_j| \leq 2) \leq C_\alpha c_j^\alpha, \quad \mathbb{E}|w(c_j X_j)| \geq c_\alpha c_j^\alpha. \quad (94)$$

The gates depend on independent input coordinates. If N_{open} is the number of open gates, then $\mathbb{E}[N_{\text{open}}(N_{\text{open}} - 1)] \leq C_\alpha F^2$. Before the outer truncation, each summand in (93) has mean zero because its gate variable is independent of its receiver variables. The truncation changes the sum only when at least two gates are open, so

$$|\mathbb{E} \Psi_B(\mathbf{X})| \leq C_\alpha F^2. \quad (95)$$

It remains to control the same statistic at $\mathbf{Y} = (I + B)\mathbf{X}$. For fixed $0 < \delta < 1/4$, (89) and the stable tail bound imply

$$\mathbb{P}\{\exists i : |c_i(B\mathbf{X})_i| > \delta\} \leq C_\alpha \delta^{-\alpha} \sum_i c_i^\alpha \omega_i(B) \leq C_\alpha \delta^{-\alpha} F^2. \quad (96)$$

On the complementary event, Lipschitz continuity of w and (94) show that replacing every output gate $w(c_j Y_j)$ by its input gate $w(c_j X_j)$ changes the expectation by at most

$$C_\alpha \delta F + C_\alpha \delta^{-\alpha} F^2. \quad (97)$$

After this replacement, removing the outer truncation costs at most another $C_\alpha F^2$.

Finally, the random vector $B(I - e_j \otimes e_j)\mathbf{X}$ is independent of the gate variable X_j , and (91) shows that replacing $\Theta_j(\mathbf{Y})$ by $\Theta_j(\mathbf{X} + c_j X_j u_j)$ costs, after summing over j , at most

$$C_\alpha F \tau_\alpha(F). \quad (98)$$

Because $(u_j)_j = 0$, conditioning on X_j and using Equations (92) and (94) gives

$$\sum_j \mathbb{E}[w(c_j X_j) \chi(\Theta_j(\mathbf{X} + c_j X_j u_j))] \geq c_\alpha F. \quad (99)$$

Choose δ and then F sufficiently small. Equations (95) and (97) to (99) give

$$\mathbb{E}\Psi_B(\mathbf{Y}) - \mathbb{E}\Psi_B(\mathbf{X}) \geq c_\alpha F.$$

The test can be nonzero only when a gate is open. The coordinates of $\mathbf{Y}$ have stable scales bounded by $(1 + F)^{1/\alpha}$, so

$$\mathbb{E}\Psi_B(\mathbf{X})^2 + \mathbb{E}\Psi_B(\mathbf{Y})^2 \leq C_\alpha F.$$

The Hellinger Cauchy–Schwarz inequality now gives $h^2((I + B)_\# \mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha F$, which is (88).

We use this estimate together with uniform invariance under finite tail sign changes, thereby avoiding any triangularity assumption. Let B have ℓ^α rows, zero diagonal, $\|B\|_{\text{op}} \leq 1/10$, and belong to $\mathcal{S}_2$. Suppose $I + B$ admits a compatible stable row-series realization and $(I + B)_\# \mu_\alpha \sim \mu_\alpha$. For a finite coordinate set $\mathcal{J}$, let $Q_\mathcal{J}$ change the signs in $\mathcal{J}$ and set

$$\Delta_\mathcal{J} = Q_\mathcal{J} B Q_\mathcal{J} - B, \quad A_\mathcal{J} = (I + B)^{-1} Q_\mathcal{J} (I + B) Q_\mathcal{J}.$$

The matrix $\Delta_\mathcal{J}$ contains precisely the entries crossing the cut between $\mathcal{J}$ and its complement. Because $\mathcal{J}$ is finite, these entries form a sum of finitely many ℓ^2 columns and finitely many ℓ^α rows; hence $F_\alpha(\Delta_\mathcal{J}) < \infty$. The finite-dimensional bound (88) extends to such infinite matrices by applying it to $P_n \Delta_\mathcal{J} P_n$ and using the Hellinger convergence from the already proved sufficiency argument.

Put

$$K_\mathcal{J} = (I + B)^{-1} \Delta_\mathcal{J}, \quad D_\mathcal{J} = I + \text{diag } K_\mathcal{J}, \quad \tilde{B}_\mathcal{J} = D_\mathcal{J}^{-1} (I + K_\mathcal{J}) - I.$$

Then $A_\mathcal{J} = I + K_\mathcal{J}$ and $\text{diag } \tilde{B}_\mathcal{J} = 0$. Since $\|(I + B)^{-1} - I\|_{\text{op}} \leq 1/9$, orthogonal projection away from the diagonal entry in each column gives

$$\frac{8}{9} \|\Delta_\mathcal{J} e_j\|_2 \leq \|(I - e_j \otimes e_j) K_\mathcal{J} e_j\|_2 \leq \frac{10}{9} \|\Delta_\mathcal{J} e_j\|_2.$$

For small $F_\alpha(\Delta_\mathcal{J})$, the diagonal entries of $D_\mathcal{J}$ lie in $[1/2, 3/2]$, and consequently

$$F_\alpha(\tilde{B}_\mathcal{J}) \asymp_\alpha F_\alpha(\Delta_\mathcal{J}), \quad \|\text{diag } K_\mathcal{J}\|_2 \leq \frac{1}{9} F_\alpha(\Delta_\mathcal{J})^{1/\alpha}. \quad (100)$$

The lower bound for $\tilde{B}_{\mathcal{J}}$, the diagonal scale upper bound for $D_{\mathcal{J}}$, and the Hellinger triangle inequality now give

$$F_{\alpha}(\Delta_{\mathcal{J}}) \leq \varepsilon_* \implies h((A_{\mathcal{J}})_{\#}\mu_{\alpha}, \mu_{\alpha}) \geq c_* F_{\alpha}(\Delta_{\mathcal{J}})^{1/2}. \quad (101)$$

Indeed, by (88), the two terms have respective orders $F_{\alpha}(\Delta_{\mathcal{J}})^{1/2}$ and $F_{\alpha}(\Delta_{\mathcal{J}})^{1/\alpha}$, and the latter is smaller for small cost because $\alpha < 2$.

On the other hand, finite-prefix approximation of the likelihood ratio gives uniform invariance under finite tail sign changes:

$$\sup_{\substack{\mathcal{J} \subset \{N+1, N+2, \dots\} \\ |\mathcal{J}| < \infty}} h((I+B)_{\#}\mu_{\alpha}, (Q_{\mathcal{J}})_{\#}(I+B)_{\#}\mu_{\alpha}) \longrightarrow 0. \quad (102)$$

Hellinger invariance and $(Q_{\mathcal{J}})_{\#}\mu_{\alpha} = \mu_{\alpha}$ also give the connecting identity

$$h((I+B)_{\#}\mu_{\alpha}, (Q_{\mathcal{J}})_{\#}(I+B)_{\#}\mu_{\alpha}) = h(\mu_{\alpha}, (A_{\mathcal{J}})_{\#}\mu_{\alpha}). \quad (103)$$

If $F_{\alpha}(B) = \infty$, deleting finitely many columns changes the cost by a finite amount, and so does deleting finitely many rows because those rows lie in ℓ^{α} . Hence every tail still has infinite column cost. Let I_0 be a finite coordinate set in such a tail, and form $\mathcal{J} \subset I_0$ by independently including each coordinate with probability one half. Column by column, concavity of $t^{\alpha/2}$ gives

$$\mathbb{E}_{\mathcal{J}} F_{\alpha}(P_{I_0} \Delta_{\mathcal{J}} P_{I_0}) \geq 2^{\alpha-1} F_{\alpha}(P_{I_0} B P_{I_0}). \quad (104)$$

As I_0 increases through finite subsets of the tail, the right side is unbounded, so some finite cuts have arbitrarily large cost. Equivalence also makes every sufficiently remote one-coordinate output marginal close to ρ_{α} , so

$$\omega_i(B) := \sum_{j \neq i} |b_{ij}|^{\alpha} \longrightarrow 0.$$

Together with $B \in \mathcal{S}_2$, which gives $c_i := \|B e_i\|_2 \rightarrow 0$, this yields

$$F_{\alpha}(\Delta_{\{i\}}) = 2^{\alpha}(c_i^{\alpha} + \omega_i(B)) \longrightarrow 0.$$

For $\alpha \leq 1$, F_{α} is subadditive; for $\alpha \geq 1$, $F_{\alpha}^{1/\alpha}$ is the $\ell^{\alpha}(\ell^2)$ norm and satisfies the triangle inequality. Moreover, when $i \notin \mathcal{J}$,

$$\Delta_{\mathcal{J} \cup \{i\}} - \Delta_{\mathcal{J}} = Q_{\mathcal{J}} \Delta_{\{i\}} Q_{\mathcal{J}},$$

and sign multiplication preserves the relevant subadditive size. Thus the increment from adding one sufficiently remote coordinate tends uniformly to zero in that size. Fix $0 < \varepsilon < \varepsilon_*/2$. Starting with the empty cut and adding the elements of a finite cut supplied by (104), stop the first time the corresponding threshold is crossed. For every sufficiently remote tail this produces a finite $\mathcal{J}$ with $\varepsilon \leq F_{\alpha}(\Delta_{\mathcal{J}}) \leq 2\varepsilon$. Equations (101) to (103) then contradict one another. Hence $F_{\alpha}(B) < \infty$.

Finally, put $M = I + K$ in (87) and let $K_n = P_n K P_n$ and $M_n = I + K_n$. For a sufficiently large n , the equivalent transformation $A = M_n^{-1} M$ is arbitrarily close to the identity in operator norm. Indeed, M_n changes only finitely many coordinates, so $(M_n)_{\#}\mu_{\alpha} \sim \mu_{\alpha}$; compatible composition with $M_{\#}\mu_{\alpha} \sim \mu_{\alpha}$ gives $A_{\#}\mu_{\alpha} \sim \mu_{\alpha}$. Its diagonal part $D = \text{diag } A$ is positive, boundedly

invertible, and $D - I \in \mathcal{S}_2$; the one-coordinate scale criterion therefore gives $D_{\#}\mu_{\alpha} \sim \mu_{\alpha}$. Consequently

$$D^{-1}A = I + B, \quad \text{diag } B = 0, \quad B \in \mathcal{S}_2, \quad \|B\|_{\text{op}} \leq 1/10, \quad (I + B)_{\#}\mu_{\alpha} \sim \mu_{\alpha}.$$

Finite-coordinate and bounded diagonal composition preserve the ℓ^{α} row condition and the compatible stable row-series realization. Thus the normalized B satisfies the hypotheses of the preceding finite-cut contradiction, which gives $F_{\alpha}(B) < \infty$. Since K_n is supported on a finite coordinate block, we have

$$M = M_n D(I + B) = D + (K_n D + DB + K_n DB).$$

The term in parentheses has finite column cost by (79) and finite-coordinate support. Restoring the initial signed permutation therefore gives

$$\sum_j \left(\sum_{i \neq j} |k_{ij}|^2 \right)^{\alpha/2} < \infty,$$

while $K \in \mathcal{S}_2$ already gives $\sum_i |k_{ii}|^2 < \infty$. This is precisely $\mathfrak{e}_{\alpha}(TQ^{-1}) < \infty$, and completes the necessity.

If $\nu_{\alpha,T} = \mu_{\alpha}$, comparison of the joint characteristic functions gives $\|T^*u\|_{\alpha} = \|u\|_{\alpha}$ for every finitely supported u . The same statement for the compatible inverse makes T^* a surjective isometry of ℓ^{α} . For $0 < \alpha < 2$, the equality case of

$$2\|u\|_{\alpha}^{\alpha} + 2\|v\|_{\alpha}^{\alpha} - \|u + v\|_{\alpha}^{\alpha} - \|u - v\|_{\alpha}^{\alpha} \geq 0$$

shows that disjointly supported vectors have disjointly supported images. The images of the coordinate vectors therefore have pairwise disjoint supports; surjectivity makes each support a singleton, and norm preservation makes its nonzero entry ± 1 . Thus $T \in \mathcal{P}^{\pm}$, while every member of $\mathcal{P}^{\pm}$ plainly preserves μ_{α} . This proves the assertion about the exact linear stabilizer and completes the proof. $\square$

4.1 Anisotropic stable Hellinger coercivity and sharpness

The two summands in (74) have genuinely different homogeneities. Let $\mathcal{P}_n^{\pm}$ denote the group of all $n \times n$ signed permutation matrices. For a finite matrix A , put

$$\mathfrak{e}_{\alpha,n}(A) = \min_{Q \in \mathcal{P}_n^{\pm}} \mathfrak{e}_{\alpha}(AQ^{-1}).$$

Proposition 4.2 (Anisotropic stable Hellinger coercivity). *Let $A \in \text{GL}(n, \mathbb{R})$. If all singular values of A lie in a fixed interval $[a, b] \subset (0, \infty)$, then constants depending only on α, a, b , and not on the dimension, satisfy*

$$c_{\alpha,a,b} \min\{\mathfrak{e}_{\alpha,n}(A), 1\} \leq h^2(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}) \leq C_{\alpha,a,b} \min\{\mathfrak{e}_{\alpha,n}(A), 1\}. \quad (105)$$

Proof. The upper bound is (81), after removing the minimizing signed permutation. For the even convolution power k constructed in the proof of theorem 4.1, the finite-dimensional Fourier transform gives the quantitative inequality

$$h^2(((A^{-1})^*)_{\#}(\lambda^{*k})^{\otimes n}, (\lambda^{*k})^{\otimes n}) \leq k h^2(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}). \quad (106)$$

Indeed, the unitary Fourier transform intertwines the half-density actions of A and $(A^{-1})^*$; taking moduli is an L^2 contraction. For k tensor copies, squared Hellinger distance grows by at most the factor k , and coordinatewise frequency addition is a data-processing map. Apply the finite-variance coercivity theorem to the product of the λ^{*k} law rescaled to variance one. When the singular values of A lie in $[a, b]$, those of $(A^{-1})^*$ lie in $[b^{-1}, a^{-1}]$, and

$$A - Q = -A((A^{-1})^* - Q)^*Q$$

shows that distance from the signed-permutation group changes by at most a factor depending on a, b . Thus (106) gives

$$h^2(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_{\alpha,a,b} \min \left\{ \min_{Q \in \mathcal{P}_n^\pm} \|A - Q\|_F^2, 1 \right\}. \quad (107)$$

Near a signed permutation, remove that permutation and write

$$d_{\text{diag}} = \sum_i |a_{ii} - 1|^2, \quad F = \sum_j \left(\sum_{i \neq j} |a_{ij}|^2 \right)^{\alpha/2}.$$

For $d_{\text{diag}} + F$ small, the identity is the nearest signed permutation, so (107) controls d_{diag} . If $D = \text{diag } A$ and $D^{-1}A = I + B$, then $\text{diag } B = 0$ and $F_\alpha(B) \asymp_\alpha F$. The off-diagonal column lower bound and the diagonal scale upper bound give

$$h(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha \sqrt{F} - C_\alpha \sqrt{d_{\text{diag}}}.$$

If d_{diag} dominates F , use (107); otherwise absorb the second term in the last display. This proves the local lower bound in (105). If uniform separation failed away from that neighborhood, one could take a block-diagonal sum of counterexamples whose squared Hellinger distances are summable. Kakutani's theorem would make the resulting infinite stable products equivalent. The Hilbert–Schmidt part of [theorem 4.1](#) forces the signed permutation furnished by that theorem to preserve every sufficiently late block; otherwise each crossed block contributes a fixed positive square error. The restrictions of that permutation to the late blocks would then make their anisotropic mixing costs summable, a contradiction. This proves the global lower bound. $\square$

The different local powers are already visible in two dimensions. For the shear $T_t(x, y) = (x + ty, y)$, conditioning on y and applying Plancherel gives the exact formula

$$h^2((T_t)_{\#}\mu_{\alpha,2}, \mu_{\alpha,2}) = 2 \int_{\mathbb{R}} (1 - e^{-|t\xi|^\alpha}) |\widehat{g}_\alpha(\xi)|^2 d\xi. \quad (108)$$

Consequently

$$h^2((T_t)_{\#}\mu_{\alpha,2}, \mu_{\alpha,2}) \sim 2|t|^\alpha \int_{\mathbb{R}} |\xi|^\alpha |\widehat{g}_\alpha(\xi)|^2 d\xi, \quad t \rightarrow 0,$$

whereas a diagonal scale perturbation has squared Hellinger cost of order t^2 . Thus anisotropic stable Hellinger coercivity is not generated by a quadratic form from one finite-dimensional matrix Fisher operator.

In particular, Hilbert–Schmidt proximity is necessary but no longer sufficient. On independent two-coordinate blocks, let

$$T_a(x_k, y_k)_{k \geq 1} = (x_k + a_k y_k, y_k)_{k \geq 1}.$$

Then $T_a - I \in \mathcal{S}_2$ exactly when $a \in \ell^2$, while [theorem 4.1](#) gives $(T_a)_\# \mu_\alpha \sim \mu_\alpha$ exactly when $a \in \ell^\alpha$. Any $a \in \ell^2 \setminus \ell^\alpha$ therefore gives a Hilbert–Schmidt perturbation with singular image law, even though the stable coordinate density is everywhere positive and has finite location and scale Fisher information. Conversely, when one source is mixed into many outputs, the criterion charges the ℓ^2 -norm of that whole column only once. An entrywise ℓ^α condition would therefore be too strong. The correct stable criterion is the anisotropic mixing cost [\(74\)](#), not a Schatten-class replacement for the finite-variance theorem.

5 Zero-Set Row Condition and Support Geometry

Positivity in the main theorem removes a separate null-set obstruction from the finite-marginal likelihood ratios. When the density has zeros, Hilbert–Schmidt proximity remains the metric accumulation condition, but equivalence also requires the zero-set row condition for the operator and its inverse. We first prove this exact replacement, then show how bounded and one-sided supports can reduce the equivalence group to permutation classes. The hypotheses are stated separately for each result; [\(3\)](#) is not a standing assumption.

5.1 The metric input when the density has zeros

The Fourier/Fisher mechanism still supplies the metric part of the classification; for its matrix estimates the appropriate regularity is

$$\begin{aligned} f \geq 0, \quad \int f = 1, \quad g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \quad xg' \in L^2(\mathbb{R}), \\ \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad f \text{ non-Gaussian.} \end{aligned} \tag{109}$$

Here g is defined on the whole real line, including by zero outside the support. Regularity only in the interior of the support is insufficient. For example, the uniform density has zero interior derivative but its square root has jumps at the boundary.

Proposition 5.1 (Local Fourier coercivity with zeros of the density). *Under [\(109\)](#), the matrix Fisher operator decomposition, finite-dimensional Hellinger path bound, and dimension-free local Frobenius coercivity in [Section 2](#) remain valid. Write $\mu = \rho^{\otimes \mathbb{N}}$, $\rho(dx) = f(x)dx$. There are constants $c, C, \delta > 0$, with $\delta < 1$, such that*

$$c\|K\|_{\mathcal{S}_2} \leq h(\mu, \nu_{I+K}) \leq C\|K\|_{\mathcal{S}_2} \quad (\|K\|_{\mathcal{S}_2} \leq \delta). \tag{110}$$

Moreover, for every $T \in \text{GL}(H)$, the following necessary condition survives:

$$\nu_T \ll \mu \implies T - Q \in \mathcal{S}_2 \text{ for some } Q \in \mathcal{P}. \tag{111}$$

For orthogonal transformations one may omit $xg' \in L^2$, retaining non-Gaussianity, $g \in W^{1,2}(\mathbb{R})$, and normalization to a centered law with variance one. These statements do not assert absolute continuity of ν_{I+K} .

Proof. A Sobolev function has weak derivative zero almost everywhere on its zero set. Define $s = -2g'/g$ on $\{g > 0\}$ and $s = 0$ elsewhere. Then $sg = -2g'$ almost everywhere on $\mathbb{R}$. The cutoff integration-by-parts identities, including $\mathbb{E}s(X) = 0$, $\mathbb{E}[Xs(X)] = 1$, and the formulas for J, κ , are therefore unchanged. Equality $J = 1$ still gives the global equation $g' = -xg/2$; it cannot

produce a Gaussian truncated at a boundary. Similarly, $\kappa = 0$ would give $2xg' + g = 0$ globally and no nonzero square-integrable probability half-density. The finite real Fourier sums with dual moments and estimates for half-density derivatives use these identities and L^2 regularity, so their proofs apply.

The infinite-dimensional limiting step needs a different interpretation. For $K \in \mathcal{S}_2$ with $I + K$ invertible, put $T_n = I + P_n K P_n$. The dimension-free Hellinger path bound and invariance of Hellinger distance give, for sufficiently large m, n ,

$$h(\nu_{T_n}, \nu_{T_m}) \leq C \|T_n^{-1}\|_{\text{op}} \|T_n - T_m\|_{\mathcal{S}_2}.$$

Choose a single probability measure dominating the countable family $\mu, \nu_{T_1}, \nu_{T_2}, \dots$. Their square-root densities are Cauchy in its L^2 space. The nonnegative limit has norm one and defines a probability measure; convergence of finite output row sums identifies this measure with ν_{I+K} . This proves Hellinger convergence, not a density relative to μ . Passing the finite local bounds to this limit proves (110). For orthogonal operators, use the finite-coordinate polar approximants and skew-generator bound from [proposition 3.5](#) in the same way.

Finally, the proof of [lemma 3.11](#) uses absolute continuity only through [lemmas 3.9](#) and [3.10](#), which assume $\nu_T \ll \mu$. Its other inputs are the moment-only characteristic-function rigidity, the measurable inverse action, and the local estimate for the finite-rank conjugated tail perturbations used in that proof. The estimate just established supplies this last input without any quasi-invariance assumption. The same tail matching and permutation argument therefore proves (111). $\square$

For a Gamma density centered with variance one, the zero-extended square root is in $W^{1,2}(\mathbb{R})$ exactly when its shape parameter exceeds two. For an affine rescaling of a Beta density, the corresponding condition is that both shape parameters exceed two. These assertions follow by squaring the derivative of the corresponding boundary power. At shape one the square root instead has a jump, so its vanishing interior power derivative is not an exception. The support classifications below require none of this regularity and include every positive Gamma or Beta shape.

5.2 The linear equivalence group for arbitrary zero sets of the density

Hilbert–Schmidt proximity remains the metric condition. The zero-set row condition adds the exact null-set requirement for T and T^{-1} , using the positivity set of the density rather than its closed support.

Theorem 5.2 (The linear equivalence group with density zeros). *Assume (109), let $\rho(dx) = f(x)dx$, and put $\mathcal{U} = \{x \in \mathbb{R} : f(x) > 0\}$. For every $T \in \text{GL}(H)$, the almost-sure requirements below are called the forward and inverse zero-set row conditions.*

$$\nu_T \ll \mu \iff \begin{cases} T - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}, \\ \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i. \end{cases} \quad (112)$$

Consequently the linear equivalence group of μ is

$$\begin{aligned} \mathcal{G}_\rho &:= \{T \in \text{GL}(H) : \nu_T \sim \mu\} \\ &= \left\{ T \in \text{GL}(H) : \begin{aligned} &T - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}, \\ &\mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0, \\ &\mu \left\{ x : \sum_j (T^{-1})_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i \end{aligned} \right\}. \end{aligned} \quad (113)$$

This set is a subgroup of $\text{GL}(H)$. For orthogonal members of $\mathcal{G}_\rho$, the same criterion holds with $T^{-1} = T^*$ and without the scale condition $xg' \in L^2$.

More precisely, whenever $T - Q \in \mathcal{S}_2(H)$ for some $Q \in \mathcal{P}$, the Lebesgue decomposition of ν_T relative to μ is its restriction to $\mathcal{U}^\mathbb{N}$ plus its restriction to the complement: the former is absolutely continuous and the latter is singular.

Proof. First suppose $T - Q \in \mathcal{S}_2$ and put $K = TQ^{-1} - I$. The exact invariance $\nu_Q = \mu$ gives $\nu_T = \nu_{I+K}$. For the eventually invertible finite-coordinate approximants $T_n = I + P_n K P_n$, the proof of [proposition 5.1](#) gives

$$h(\nu_{T_n}, \nu_T) \longrightarrow 0.$$

In particular their probabilities converge uniformly over measurable sets, by the Hellinger bound on total variation.

For each such n , the law ν_{T_n} has a Lebesgue density in the first n coordinates and retains the independent product law on the remaining coordinates. On $\mathcal{U}^n$, the density of μ_n is strictly positive. Fubini's theorem therefore gives

$$\nu_{T_n}|_{\mathcal{U}^n} \ll \mu.$$

If B is any μ -null Borel set, then $\nu_{T_n}(B \cap \mathcal{U}^\mathbb{N}) = 0$ for all sufficiently large n . Total variation convergence implies $\nu_T(B \cap \mathcal{U}^\mathbb{N}) = 0$. Thus $\nu_T|_{\mathcal{U}^\mathbb{N}} \ll \mu$. Since $\mu(\mathcal{U}^\mathbb{N}) = 1$, the restriction to its complement is singular. This proves the asserted decomposition and

$$\nu_T \ll \mu \iff \nu_T(\mathcal{U}^\mathbb{N}) = 1 \iff \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i.$$

The necessary Hilbert–Schmidt condition in [\(111\)](#) now proves [\(112\)](#) in both directions.

If $\nu_T \sim \mu$, the measurable inverse action gives $\nu_{T^{-1}} \sim \mu$, so both zero-set row conditions in [\(113\)](#) are necessary. Conversely, $T - Q \in \mathcal{S}_2$ implies

$$T^{-1} - Q^{-1} = -T^{-1}(T - Q)Q^{-1} \in \mathcal{S}_2.$$

Apply [\(112\)](#) to T and T^{-1} to obtain $\nu_T \ll \mu$ and $\nu_{T^{-1}} \ll \mu$. The measurable inverse action then gives

$$\mu = (\Phi_T)_\# \nu_{T^{-1}} \ll (\Phi_T)_\# \mu = \nu_T,$$

which proves equivalence. For $S, T \in \mathcal{G}_\rho$, measurable composition yields $\nu_{ST} = (\Phi_S)_\# \nu_T \sim \nu_S \sim \mu$; the inverse argument already proved gives $T^{-1} \in \mathcal{G}_\rho$. Thus the displayed operator class is a group. For orthogonal transformations, repeat the restriction argument with the finite-coordinate orthogonal approximants from [proposition 3.5](#); their Hellinger convergence needs only location Fisher information. $\square$

Every nonzero row sum has an absolutely continuous scalar law, so the condition is unchanged by modifying f on a null set. It is automatic when $f > 0$ almost everywhere, while the contraction in [Section 5.5](#) satisfies only the forward condition and therefore gives strict absolute continuity rather than equivalence.

5.3 Boundary behavior of the density and summability criteria

The zero-set row condition governs absolute continuity, whereas projected Hellinger energy governs accumulated overlap. Hard and smoothly vanishing boundaries separate these effects.

Let ρ be uniform on $[-\sqrt{3}, \sqrt{3}]$, let $\mu_2 = \rho^{\otimes 2}$, and consider $T = \bigoplus_k O_{\theta_k}$. Write $\delta_k = \min_{m \in \mathbb{Z}} |\theta_k - m\pi/2| \in [0, \pi/4]$. For one block the affinity is the relative area of intersection of a square and its rotation. Removing the four corner triangles gives, for $0 < \delta \leq \pi/4$,

$$\begin{aligned} A(\mu_2, (O_\delta)_\# \mu_2) &= 1 - \frac{(\cos \delta + \sin \delta - 1)^2}{2 \sin \delta \cos \delta}, \\ 1 - A(\mu_2, (O_\delta)_\# \mu_2) &= \frac{\delta}{2} + O(\delta^2). \end{aligned} \tag{114}$$

At $\delta = 0$, the affinity is one. Each nonzero δ leaves positive area on both sides of the support difference. Consequently [theorem 5.4](#) gives the complete trichotomy

$$\begin{array}{l|l} \delta_k = 0 \text{ for every } k & \nu_T = \mu \\ 0 < \sum_k \delta_k < \infty & \nu_T \not\ll \mu, \quad \nu_T \not\ll \mu, \quad \mu \not\ll \nu_T \\ \sum_k \delta_k = \infty & \nu_T \perp \mu. \end{array} \tag{115}$$

In particular, $\theta_k = 1/k$ gives $T - I \in \mathcal{S}_2$ but $\nu_T \perp \mu$. With $\theta_k = 1/k^2$, the laws instead have positive affinity and neither is absolutely continuous with respect to the other.

For comparison, rescale to variance one a density proportional to $(1 - x^2)^r \mathbf{1}_{\{|x| < 1\}}$, $r > 1$. Its zero-extended square root satisfies the Sobolev hypotheses. Let f be this density, $\rho(dx) = f(x) dx$, $\mu_2 = \rho^{\otimes 2}$, and $g = \sqrt{f}$. Writing $J = 4\|g'\|_2^2 > 1$, the skew-generator formula gives

$$1 - A(\mu_2, (O_\theta)_\# \mu_2) = \frac{J-1}{4} \theta^2 + o(\theta^2). \tag{116}$$

Indeed, for the planar skew generator $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, one has $\|A\|_F^2 = 2$ and $h^2 = \theta^2 \langle A, \mathcal{I}A \rangle_F / 4 + o(\theta^2)$. The only rotations preserving the square support are its signed permutation symmetries. Every rotated square shares a neighborhood of the origin with the original square, on which both densities are positive; hence $A(\mu_2, (O_\theta)_\# \mu_2) > 0$ for every angle. Affinity is continuous and strictly below one away from them, so the three cases in [\(115\)](#) now use $\sum_k \delta_k^2$ in place of $\sum_k \delta_k$. Equivalence still requires every $\delta_k = 0$. Thus the quadratic metric supplied by the Fourier/Fisher method controls accumulation of Hellinger loss, while the zero-set row condition separately controls absolute continuity.

5.4 Complete linear classification for bounded and half-line laws

Write $\rho = \text{Law}(X)$, $\rho^- = \text{Law}(-X)$, and $\mu = \rho^{\otimes \mathbb{N}}$. Throughout this subsection X is centered with variance one. The operator action is the measurable row-series realization of [lemma 3.1](#).

For bounded and one-sided supports, conditional expectations turn the support boundary into an ℓ^1 -type row constraint, and the inverse relations force a monomial operator. No density or Fisher information is needed.

Theorem 5.3 (Linear equivalence for bounded and half-line coordinate laws). *Let $T \in \text{GL}(H)$.*

1. *Suppose $a = \text{ess inf } X < 0 < b = \text{ess sup } X$ are finite. Then $\nu_T \sim \mu$ forces T to be a signed permutation. The complete equivalence group is as follows:*

<i>relation between ρ^- and ρ</i>	<i>allowed signed permutations</i>
$\rho^- = \rho$	<i>all signs</i>
$\rho^- \sim \rho, \rho^- \neq \rho$	<i>only finitely many negative signs</i>
$\rho^- \not\sim \rho$	<i>no negative signs.</i>

In particular, $a + b \neq 0$ permits only ordinary permutations. The support need not fill an interval, and a density is not required.

2. *Suppose $\text{ess inf } X = a > -\infty$ and $\text{ess sup } X = +\infty$. Then*

$$\nu_T \sim \mu \iff T \text{ is an ordinary coordinate permutation.} \quad (117)$$

The same conclusion holds for a law bounded above and unbounded below.

In the second case every equivalence is exact invariance. Failure of any of these equivalence criteria does not, by itself, imply singularity.

Proof. Assume $\nu_T \sim \mu$, and set $S = T^{-1}$. The measurable inverse action implies $(\Phi_S)_\# \nu_T = \mu$. Pushing equivalence forward by Φ_S therefore gives $\nu_S \sim \mu$. For any row v of T or S , the random series $Z = \sum_j v_j X_j$ converges in L^2 . Centering and independence imply

$$\mathbb{E}[Z \mid X_1, \dots, X_m] = \sum_{j \leq m} v_j X_j. \quad (118)$$

In the bounded case $Z \in [a, b]$ almost surely. Its conditional expectation lies in the same interval. Independence and the definition of essential endpoints show that the essential width of the finite sum on the right of (118) is $(b - a) \sum_{j \leq m} |v_j|$. Consequently every row of both T and S has ℓ^1 -norm at most one. This proves the required bound even when the original row has infinitely many nonzero entries.

Let $A_{ij} = |T_{ij}|$ and $B_{ij} = |S_{ij}|$. The products defining the entries of TS and ST are absolutely convergent by the ℓ^2 row/column Cauchy–Schwarz inequality. Thus $AB \geq I$ and $BA \geq I$ entrywise. Tonelli’s theorem and the row bounds give

$$\sum_j (AB)_{ij} = \sum_k A_{ik} \sum_j B_{kj} \leq 1, \quad \sum_j (BA)_{ij} \leq 1.$$

It follows that $AB = BA = I$. This reasoning concerns nonnegative matrix entries and does not assume that entrywise absolute values define bounded operators on ℓ^2 .

For completeness, nonnegative matrices inverse to one another in this sense are monomial. For each i , choose k with $A_{ik}B_{ki} > 0$. The off-diagonal entries of $AB = I$ force row k of B to vanish outside column i . Then $BA = I$ forces row i of A to vanish outside column k . The resulting correspondence is bijective. Its weights and their reciprocals are at most one by the row bounds, so all weights are one. Hence T is a signed permutation.

For a negative row sign, scalar equivalence requires $\rho^- \sim \rho$, which in particular requires $a = -b$. The remaining classification is a product calculation. When these laws are equivalent,

$A(\rho, \rho^-) > 0$, and $A(\rho, \rho^-) = 1$ exactly when they are equal. Each negative sign contributes the factor $A(\rho, \rho^-)$, while positive signs contribute one. Kakutani's theorem [10] gives precisely the three alternatives stated above.

For the lower-bounded, upper-unbounded case, centering and nondegeneracy give $a < 0$. Now every finite sum in (118) is bounded below by a . A negative coefficient would make its essential infimum $-\infty$, which is impossible. Thus every $v_j \geq 0$, and its finite essential infimum gives

$$a \sum_{j \leq m} v_j \geq a, \quad \text{hence} \quad \sum_j v_j \leq 1.$$

Both T and S are therefore nonnegative with row sums at most one. The same inverse-matrix argument makes T an ordinary permutation. Such a permutation preserves the i.i.d. product, proving the converse. Reflection proves the upper-bounded case. $\square$

For $Y \sim \text{Gamma}(r_\Gamma, \theta_\Gamma)$, where $r_\Gamma > 0$ is the shape and $\theta_\Gamma > 0$ is the scale, put

$$X = \frac{Y - r_\Gamma \theta_\Gamma}{\sqrt{r_\Gamma} \theta_\Gamma}.$$

Its essential endpoints are $-\sqrt{r_\Gamma}$ and $+\infty$. Thus (117) gives the complete homogeneous linear equivalence classification for every $r_\Gamma, \theta_\Gamma > 0$, including the shape-two Fourier example in Section 2.4: only ordinary coordinate permutations are allowed. Compact support permits signs only to the extent allowed by the scalar pair ρ, ρ^- . For compactly supported densities positive almost everywhere inside $(-a, a)$, reflection equivalence is automatic, so an asymmetric such density permits finitely many sign changes although they do not preserve the measure exactly. An asymmetric Beta law centered and rescaled to variance one has unequal endpoint magnitudes, so it permits only permutations. These conclusions come from support geometry rather than from the Fourier/Fisher estimates.

5.5 Why the zero-set row condition is indispensable

Even a rank-one perturbation separates positive Hellinger overlap from two-sided absolute continuity.

Let f be the rescaling to variance one of $C(1 - x^2)^2 \mathbf{1}_{\{|x| < 1\}}$, and write its support as $[-a, a]$. Set $\rho(dx) = f(x) dx$. Its zero-extended square root belongs to $W^{1,2}(\mathbb{R})$, and $x(\sqrt{f})' \in L^2(\mathbb{R})$. For $0 < c < 1$, set

$$T = c \oplus I, \quad f_c(y) = c^{-1} f(y/c), \quad \mu = \rho^{\otimes \mathbb{N}}.$$

Then $T - I$ has rank one and

$$\nu_T = (f_c(y) dy) \otimes \rho^{\otimes \{2,3,\dots\}} \ll \mu, \quad \mu \not\ll \nu_T.$$

Indeed, f_c is positive on $(-ca, ca)$ and zero on the rest of $(-a, a)$. Nevertheless,

$$A(\mu, \nu_T) = \int \sqrt{f(y) f_c(y)} dy > 0, \quad \sup_n E_n(T) < \infty.$$

Thus neither Hilbert–Schmidt proximity nor bounded projected Hellinger energy implies equivalence without the inverse zero-set row condition. The failure occurs with finite location and scale Fisher information.

5.6 Exact block criteria without equivalence of the factors

The boundary examples use a block-product criterion that separates local absolute continuity from accumulated Hellinger loss.

Theorem 5.4 (Block products without equivalence of the factors). *Let $\mu = \bigotimes_{k \geq 1} \lambda_k$ and $\nu = \bigotimes_{k \geq 1} \eta_k$ be probability products over finite-dimensional coordinate blocks. Set*

$$r_k = A(\lambda_k, \eta_k), \quad L = \sum_k -\log r_k \in [0, \infty], \quad -\log 0 = +\infty.$$

Then

$$A(\mu, \nu) = \prod_k r_k, \quad \mu \perp \nu \iff L = \infty. \quad (119)$$

If $L < \infty$, then

$$\begin{aligned} \nu \ll \mu &\iff \eta_k \ll \lambda_k \text{ for every } k, \\ \mu \ll \nu &\iff \lambda_k \ll \eta_k \text{ for every } k. \end{aligned} \quad (120)$$

Thus positive affinity allows equivalence, either direction of strict absolute continuity, or neither direction. For a linear block $A_k \in \text{GL}(d_k)$ and $\lambda_k = \rho^{\otimes d_k}$, where $\rho(dx) = f(x) dx$, the comparison density is

$$|\det A_k|^{-1} \prod_{i=1}^{d_k} f((A_k^{-1}y)_i).$$

The local absolute-continuity conditions concern its actual zero set, not just the closure of its support.

Proof. Affinity tensorizes on each finite block prefix. Convergence of projected affinities to global affinity follows by conditional expectation under the common dominating measure $(\mu + \nu)/2$; the proof is also given below. This gives (119).

Necessity of each local absolute-continuity condition follows by projection. For sufficiency, suppose $\eta_k \ll \lambda_k$ for all k and $\prod_k r_k > 0$. With $\ell_n = \prod_{k \leq n} d\eta_k/d\lambda_k$, independence gives

$$\|\sqrt{\ell_m} - \sqrt{\ell_n}\|_{L^2(\mu)}^2 = 2 \left(1 - \prod_{n < k \leq m} r_k \right) \longrightarrow 0.$$

The square roots have norm one, so their limit squared is a probability density. Cylinder tests identify its measure with ν . Interchanging μ, ν proves the reverse assertion. This is the product argument behind Kakutani's criterion; the mutual absolute-continuity hypothesis on each factor cannot be discarded [10, 13]. $\square$

Corollary 5.5 (Masses of the absolutely continuous parts). *Take $m_k = (\lambda_k + \eta_k)/2$, with densities p_k, q_k , and put*

$$a_k = \lambda_k(\{q_k > 0\}), \quad b_k = \eta_k(\{p_k > 0\}).$$

If $L < \infty$, denote the absolutely continuous parts of ν relative to μ , and of μ relative to ν , by ν_a and μ_a , respectively. Then

$$\nu_a(\mathbb{R}^{\mathbb{N}}) = \prod_k b_k, \quad \mu_a(\mathbb{R}^{\mathbb{N}}) = \prod_k a_k. \quad (121)$$

Proof. Since $r_k^2 \leq a_k b_k$ and $L < \infty$, both displayed products are positive. On $C = \prod_k \{p_k q_k > 0\}$, the two conditional product laws have equivalent factors and affinities $r_k / \sqrt{a_k b_k}$ with positive product; hence they are equivalent. Outside C , each measure is carried by a set null for the other. Their restrictions to C are therefore exactly their absolutely continuous parts. $\square$

Gamma scale changes and the effect of centering. Fix $r_\Gamma > 0$. For the uncentered shape- r_Γ , scale-one Gamma density, positive scaling by c preserves the scalar measure class and gives

$$\int_0^\infty \sqrt{f(y)c^{-1}f(y/c)} dy = \left(\frac{2\sqrt{c}}{1+c} \right)^{r_\Gamma} = \cosh\left(\frac{\log c}{2}\right)^{-r_\Gamma}. \quad (122)$$

Integration of $y^{r_\Gamma-1}e^{-(1+c^{-1})y/2}$ proves the formula. Since $\log \cosh(u/2) \sim u^2/8$ at zero, it follows that

$$\bigotimes_k \text{Law}(c_k Y) \sim \bigotimes_k \text{Law}(Y) \iff \sum_k (\log c_k)^2 < \infty, \quad c_k > 0. \quad (123)$$

The condition is equivalently $\sum_k (c_k - 1)^2 < \infty$. This does not contradict [theorem 5.3](#): for $X = (Y - r_\Gamma)/\sqrt{r_\Gamma}$, the map $Y \mapsto cY$ becomes

$$X \mapsto cX + \sqrt{r_\Gamma}(c - 1),$$

an affine map fixing the lower endpoint, not the homogeneous linear map $X \mapsto cX$.

Corollary 5.6 (Coordinatewise affine equivalence for Gamma products). *Let μ be the product of the shape- r_Γ Gamma law centered with variance one. For $T \in \text{GL}(H)$ and $b \in \ell^2$, the law of $\Phi_T(x) + b$ under μ is equivalent to μ if and only if there are a coordinate permutation π and numbers $c_i > 0$ such that*

$$(Tx)_i = c_i x_{\pi(i)}, \quad b_i = \sqrt{r_\Gamma}(c_i - 1), \quad \sum_i (\log c_i)^2 < \infty.$$

Proof. The inverse affine map has linear part T^{-1} and shift $-T^{-1}b \in \ell^2$. To justify this for the row-series versions, Cauchy–Schwarz gives $\Phi_S(z + b) = \Phi_S(z) + Sb$ wherever the selected row limits exist, for every $b \in \ell^2$. The affine orbit has second-moment bound $\|T\|_{\text{op}}^2 + \|b\|_2^2$. This translation identity and the measurable composition lemma show that the inverse affine map pushes the orbit back to μ . Its pushforward of μ is consequently also equivalent to μ . Equivalence and the lower support bound show, by conditioning finite row sums as before, that both linear parts have nonnegative entries: a negative coefficient would produce an unbounded lower tail, which no finite shift can repair. Since T and T^{-1} have nonnegative matrix entries, the off-diagonal inverse identities used above show that T is monomial with positive weights. Equality of the scalar lower endpoints forces $-\sqrt{r_\Gamma}c_i + b_i = -\sqrt{r_\Gamma}$. The product criterion (123) then proves necessity and sufficiency. $\square$

5.7 An exact finite-marginal equivalence criterion

The following finite-marginal criterion isolates the measure-theoretic content without Sobolev assumptions or a dichotomy; under (109), [theorem 5.2](#) specializes it to bounded invertible linear mixing.

Proposition 5.7 (Finite-marginal tails characterize equivalence). *Let μ, ν be any probability measures on $\mathbb{R}^{\mathbb{N}}$, and let μ_n, ν_n be their first n marginals. Choose densities p_n, q_n with respect to any common dominating measure m_n . For $M > 1$, define*

$$\text{Tail}_n(M) = \int_{\{q_n > Mp_n\}} q_n dm_n + \int_{\{p_n > Mq_n\}} p_n dm_n.$$

Then

$$\mu \sim \nu \iff \lim_{M \rightarrow \infty} \sup_n \text{Tail}_n(M) = 0. \quad (124)$$

Equivalently, every finite pair is equivalent and its forward and reverse likelihood-ratio martingales are uniformly integrable in their respective reference measures. Without any finite-marginal equivalence assumption,

$$A(\mu_n, \nu_n) \downarrow A(\mu, \nu), \quad \sup_n [-2 \log A(\mu_n, \nu_n)] < \infty \iff \mu \not\sim \nu. \quad (125)$$

Proof. If the limit in (124) is zero, a set where $p_n = 0 < q_n$ must have zero ν_n -mass for every n ; the reverse statement follows from the second integral. Hence $\mu_n \sim \nu_n$. Writing $\ell_n(x) = q_n(\pi_n x)/p_n(\pi_n x)$, the two terms in $\text{Tail}_n(M)$ are exactly

$$\mathbb{E}_\mu[\ell_n \mathbf{1}_{\{\ell_n > M\}}], \quad \mathbb{E}_\nu[\ell_n^{-1} \mathbf{1}_{\{\ell_n^{-1} > M\}}].$$

They are the two terms in $\text{Tail}_n(M)$, hence the uniform-integrability tests for the two mean-one likelihood-ratio martingales. Each L^1 limit has mass one, and cylinder tests identify the corresponding absolutely continuous measure. This gives both directions of absolute continuity. Conversely, under equivalence these martingales are the conditional expectations of the full likelihood ratio and its reciprocal, so both are uniformly integrable.

For the affinity assertion put $m = (\mu + \nu)/2$, $u = d\mu/dm$, and $v = d\nu/dm$. Both densities are bounded by two. For $\mathcal{F}_n = \sigma(x_1, \dots, x_n)$, set $u_n = \mathbb{E}_m[u \mid \mathcal{F}_n]$ and $v_n = \mathbb{E}_m[v \mid \mathcal{F}_n]$. Martingale convergence and bounded convergence imply

$$A(\mu_n, \nu_n) = \int \sqrt{u_n v_n} dm \longrightarrow \int \sqrt{uv} dm.$$

Conditional Cauchy–Schwarz gives monotonicity. Global affinity is zero exactly for mutually singular measures, which proves (125). $\square$

For bounded invertible linear mixing, finite output marginals still have Lebesgue densities, though their zero sets may differ from those of the product density. Thus (124) remains exact; theorems 5.2 and 5.3 turn it into explicit operator classifications in the regular zero-set and support-boundary regimes.

5.8 What the Finite-Variance Hypotheses Control

It remains to isolate the roles of the analytic hypotheses within the positive-density regime.

The a.e.-positive-density theorem includes the centered, variance-one Laplace density despite its cusp, as well as the normalized density proportional to e^{-x^4} ; symmetry is never used.

For examples allowing both asymmetry and heavy tails, take a density proportional to

$$(1 + y^2)^{-(\tau+1)/2} e^{\vartheta \arctan y}, \quad \tau > 2, \quad \vartheta \in \mathbb{R}, \quad (126)$$

and center and rescale to variance one. Its location score before centering and rescaling is

$$\frac{(\tau + 1)y - \vartheta}{1 + y^2}.$$

The score and its product with y are bounded, so centering and rescaling give an everywhere-positive example satisfying (3). For $2 < \tau \leq 4$ its fourth moment is infinite, and for $\vartheta \neq 0$ its two tails have unequal leading constants. Thus the theorem assumes neither evenness nor a fourth moment; $\vartheta = 0$ recovers a rescaled Student t family.

The centered Gamma example has a half-line zero set: its rotational Fourier defect detects local rotations, while the support theorem leaves only coordinate permutations for homogeneous linear equivalence. Corollary 5.6 treats the distinct affine problem, in which support-preserving scales can remain equivalent.

Support controls null-set admissibility. The second moment instead makes every ℓ^2 row a convergent random series, supplies uniform integrability of squared linear forms, and enables the Gaussian-summand extraction in lemma 3.8. Hence the first-moment ODE result has strictly broader moment scope than theorem 1.1.

Under (3), the Hellinger paths give L^1 likelihood convergence and uniform integrability. The finite Fourier witnesses use only their dimension-free $L^2(\mu_n)$ and generator bounds; tail symmetry avoids a separate mixed-law variance estimate.

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